

# Stability vs flexibility: reshaping archaeal membranes in silico

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## eLife Assessment

This **fundamental** study characterizes the mechanics and stability of bolalipids from archaeal membranes using a minimalist, physics-based computational model. The authors present a robust mesoscale model of bolalipids-containing membranes, systematically evaluating it across diverse membrane configurations. The results are **compelling**, demonstrating that the incorporation of bolalipids and regular bilayer lipids in archaeal membranes significantly enhances membrane fluidity and structural stability.

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## Abstract

Cellular membranes differ across the tree of life. In most bacteria and eukaryotes, single-headed lipids self-assemble into flexible bilayer membranes. By contrast, thermophilic archaea tend to possess bilayer lipids together with double-headed, monolayer spanning bolalipids, which are thought to enable cells to survive in harsh environments. Here, using a minimal computational model for bolalipid membranes, we explore the trade-offs at play when forming membranes. We find that flexible bolalipids form membranes that resemble bilayer membranes because they are able to assume a U-shaped conformation. Conversely, rigid bolalipids, which resemble the bolalipids with cyclic groups found in thermophilic archaea, take on a straight conformation and form membranes that are stiff and prone to pore formation when they undergo changes in shape. Strikingly, however, the inclusion of small amounts of bilayer lipids in a bolalipid membrane is enough to achieve fluid bolalipid membranes that are both stable and flexible – resolving this trade-off. Our study suggests a mechanism by which archaea can tune the material properties of their membranes as and when required to enable them to survive in harsh environments and to undergo essential membrane remodelling events like cell division.

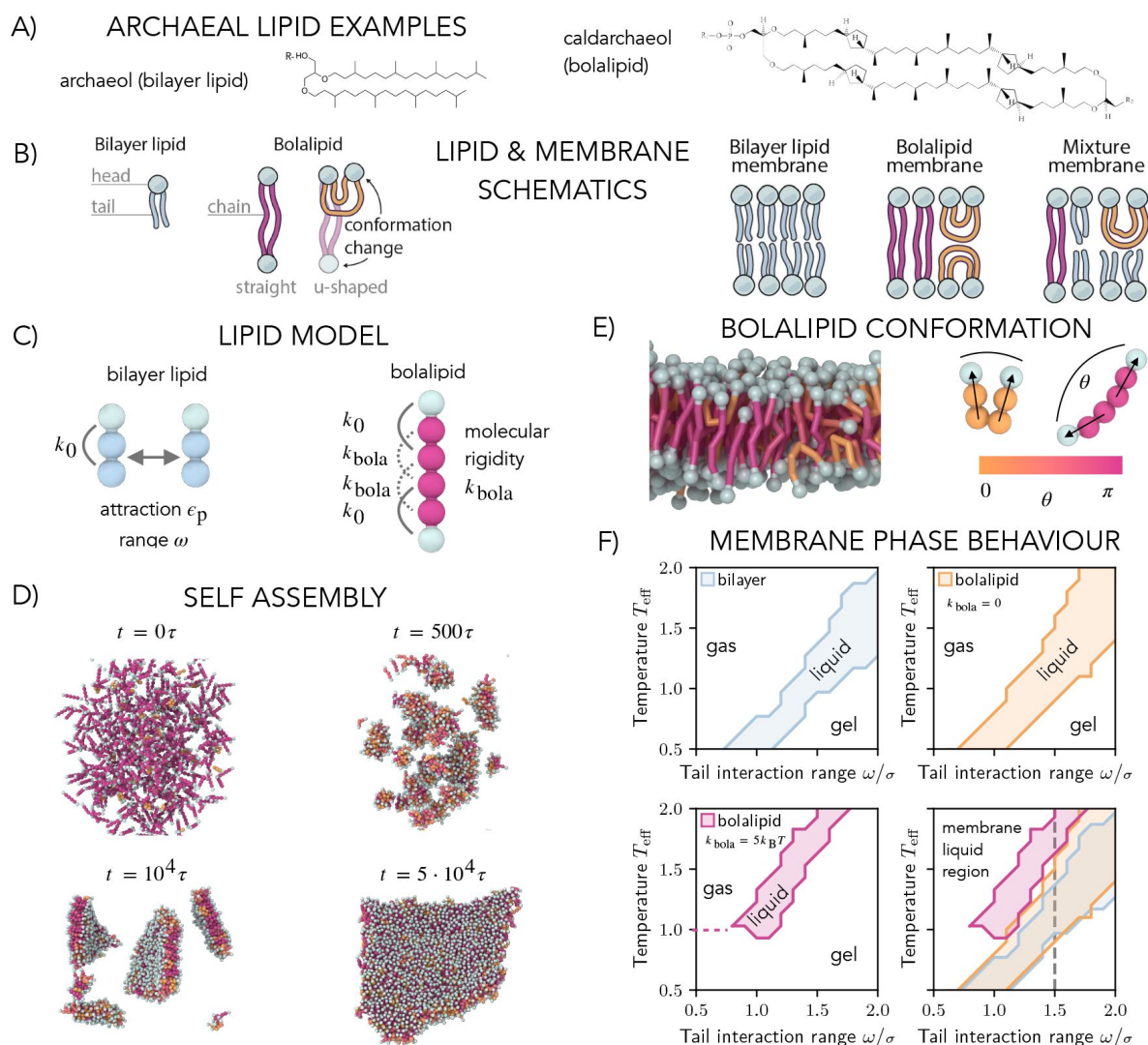
## Introduction

A bounding membrane separates the interior of every biological cell from its extracellular environment and gives the cell its shape. To perform its function, this bounding membrane must be sturdy and impermeable to the leakage of small charged molecules like ions to allow the generation of an electrochemical gradient [1], but also flexible enough to be remodelled during essential cellular processes like cell division and vesicle formation [2]. These requirements are partially conflicting and difficult to combine.

Interestingly, two different generic membrane designs have evolved across the tree of life to solve this problem [3]. Bacterial and eukaryotic membranes possess fatty acid lipids that have a hydrophilic head group linked to hydrophobic tails via ester-linkages, which self-assemble into bilayer membranes [4]. By contrast, archaeal membranes lipids are constructed from branched isoprenoid lipids. These can include cyclopentane rings and are attached via an ether linkage to one (e.g., archaeol, **Fig. 1A** left) or two hydrophilic heads, leading them to be called bipolar lipids or simply bolalipids (e.g., caldarchaeol, **Fig. 1A** right) [4]. The hydrophilic heads can be composed of different functional groups with phosphatidyl and sugar being the most relevant moieties. For bolalipids the two head groups at either end of the molecule are typically distinct (**Fig. 1A** right) [5]. As an ensemble, bilayer and bolalipids can self-assemble into fluid lipid membranes of different architectures – as bilayer lipid membranes, bolalipid membranes or mixture membranes (**Fig. 1B** right).

Like for bacteria and eukaryotes, archaea must keep their lipid membranes in a fluid state (homeoviscous adaptation). This is important even under extreme environmental conditions, such as hot and cold temperatures, or high and low pH values [7]. Because of this, many archaea adapt to changes in their environment by tuning the lipid composition of their membranes: altering the ratio between bola- and bilayer lipids in their membranes [8, 9] and/or by changing the number of cyclopentane rings in their lipid tails, which are believed to make lipid molecules more rigid [5]. For example, *Thermococcus kodakarensis* increases its tetraether bolalipid ratio from around 50% to over 80% when the temperature of the environment increases from 60 to 85°C [10]. Along the same lines, the cell membrane of *Sulfolobus acidocaldarius*, can contain over 90 % of bolalipids with up to 8 cyclopentane rings at 70°C and pH 2.5 [5, 11]. It is worth mentioning that in exceptional cases bacteria also synthesise bolalipids in response to high temperatures [12], highlighting that the study of bolalipid membranes is relevant not only for archaeal biology but also from a general membrane biophysics perspective. Besides temperature adaptation, archaeal lipids also exhibit dense packing, high viscosity, and low porosity to small molecules like protons [10].

All this makes the study of reshaping archaeal membranes extremely interesting. Unfortunately, it is fairly challenging to investigate archaeal cells experimentally due to their unusual chemistry and extreme living conditions [13]. Therefore, most of what we know about archaeal tetraether lipid membranes thus far has been collected from studying *in vitro* reconstituted membranes. For instance, the conformation of individual lipids in bolalipid membranes was studied at the water-air interface [14, 15] or using NMR experiments on lipid vesicles [16]. This suggested the existence of U-shaped lipid conformations in archaeal type membranes (**Fig. 1B**). Moreover, *in vitro* reconstituted vesicles primarily composed of bolalipids can fuse with influenza virus particles at similar kinetic rates compared to bilayer vesicles, further suggesting that bolalipids exist in U-shape allowing for membrane remodelling and fusion [17]. Membrane properties like bending stiffness [18] or lipid phase [19] have also been measured in vesicles prepared from archaeal tetraether lipids to demonstrate that archaeal lipids derived membranes are exceptional in being stable up to temperatures of 80°C [19]. Moreover, experiments with lipid vesicles made



**Figure 1.**

### Computational model and phase space of bilayer and bolalipid membranes.

(A) Structure of the diether bilayer lipid archaeol (left) and the tetraether bolalipid caldarchaeol including four cyclopentane rings (right), both present in the membrane of *Sulfolobus acidocaldarius*, a common archaeal model system that lives at high temperatures and low pH [6]. The hydrophilic head of a bolalipid can be composed of different functional groups represented by R1 and R2 (right). (B) Schematics for a bilayer lipid, a bolalipid and its two in-membrane conformations, membranes made of bilayer molecules only, bolalipid molecules only, and a mixture of the two (left to right). (C) Bilayer lipid (left) is described with one head bead and two tail beads straightened by an angular potential of strength  $k_0$ . Tail beads of different lipids attract with the strength  $\epsilon_p$  and the range  $\omega$ . Bolalipids (right) consist of two bilayer lipids connected by a bond and straightened by an angular potential of strength  $k_{bola}$ . (D) Snapshots of bolalipids self-assembling into a flat membrane ( $k_{bola} = 0.3 k_B T$ ). (E) Cross-section of self-assembled membrane (right), with bolalipids coloured according to their conformation: straight lipids in crimson and U-shaped lipids in orange. (F) Membrane phase behaviour: liquid, gel and gas regions as a function of the effective temperature  $T_{eff}$  and tail interaction range  $\omega$  for bilayer (top left) and membranes made of flexible  $k_{bola} = 0$  (top right) and stiff  $k_{bola} = 5 k_B T$  (bottom left) bolalipid molecules. Overlays of all liquid regions (bottom right) show that stiffer lipids exhibit fluid membrane region at higher temperatures. The dashed line marks  $\omega = 1.5\sigma$ , the value we used in the rest of the work.

from synthetic bilayer lipids that include cyclopentane rings, which naturally appear in lipids of extremophilic archaea, showed that increasing the number of these rings increases membrane rigidity [20].

From the view point of membrane physics, the remodelling of bilayer membranes has been studied for decades using continuum models and computer simulations [21–23]. By contrast, only a few studies have investigated bolalipid membranes applying computational or theoretical tools [24, 25]. Specifically, the pore closure time in bolalipid membranes, and the role of cyclopentane rings for membrane properties has been investigated using all-atom simulations, showing decreased lateral mobility, reduced permeability to water, and increased lipid packing [26–28]. Moreover, using coarse-grained simulations, it was suggested that bolalipid membranes are thicker [29], exhibit a gel-to-liquid phase transition at higher temperature [30], and exhibit a reduced diffusivity [31]. However, little research has been devoted to investigating mechanics and reshaping of bolalipid membranes at the mesoscale despite the obvious importance of this question from evolutionary, biophysics, and biotechnological perspectives and although different membrane physics is expected to manifest.

Here we have designed a minimal model of archaeal membranes that can be used to explore the impact of both bolalipids and bilayer lipids on membrane biophysics and membrane reshaping. Our coarse-grained molecular dynamics simulations show that the geometry of bolalipids is, through the effects of entropy alone, sufficient to shift the fluid phase of archaeal-type membranes so that they are stable at high temperatures. We show that membranes assembled from bolalipids can have a much higher bending rigidity than bilayer-derived membranes, and are more likely to resist membrane shape changes as in fission. During membrane deformation, stress in these bolalipid membranes is relieved by a small fraction of bolalipids taking up a U-shaped conformation, which renders them a mechanically switchable material. Without these U-shaped bolalipids, stress leads to the formation of large pores. However, when combined, a mixture of bilayer and bolalipids generates membranes that are stable at high temperature, which become softer as the fraction of bilayer lipids increases. Taken together, our results demonstrate how doping a bolalipid membrane with a small fraction of bilayer lipids can relieve the trade-off, enabling membranes to be both stable under extreme conditions and to be remodelled without leaking.

## Results

### Computational Model

To study bolalipid membranes and compare them to bilayers, we extended the Cooke and Deserno model for bilayer membranes [32]. In the original model for the bilayer a single bilayer lipid is represented by a chain of three nearly equally sized beads of diameter of  $\sim 1\sigma$ , where  $\sigma$  is our distance unit and roughly maps to 1 nm (Fig. 1C left); one bead stands for the head group (cyan) while the others represent the hydrophobic tail (blue). Each adjacent pair of beads in a lipid is linked by a finite extensible nonlinear elastic (FENE) bond. The angle formed by the chain of three beads is kept near  $180^\circ$  via an angular potential with strength  $k_\theta$ , instead of the approximation by a bond between end beads of the original model [32].

While lipid heads interact exclusively through volume exclusion, the beads of lipid tails interact via a soft attractive potential of the strength  $c_p$  and range  $\omega$  (Fig. 1C left), effectively modelling hydrophobic interaction in an implicit solvent. This interaction strength governs the membrane phase behaviour and can be interpreted as the effective temperature or reduced temperature  $T_{\text{eff}} = k_B T / c_p$ . As the distinction between scaling interactions ( $T_{\text{eff}}$ ) or temperature ( $T$ ) is not important for our analysis (see Supplemental Information (SI) section 14), for simplicity we refer to  $T_{\text{eff}}$  as temperature in the following.

To model a bolalipid molecule, we joined two bilayer lipids so that a lipid molecule is formed with a head bead (cyan) that is linked to four tail beads (crimson) which are again linked to another head bead (**Fig. 1C** [right](#)). In this way, both bilayer lipids and bolalipids share the same molecular structure and the same interactions between lipid beads. To decouple the effect of the connected geometry of the bolalipids from that of lipid asymmetry, we assume both head beads of a bolalipid to share the same properties. Bolalipids in archaeal membranes can differ in the number of cyclopentane rings or the branching of the tail and thus in the molecular stiffness [[7](#) [right](#), [33](#) [right](#)]. To represent this effect, we added two angular potentials between the second and the fourth and the third and the fifth tail bead with variable strength  $k_{\text{bola}}$ . By varying  $k_{\text{bola}}$ , we can control the molecular stiffness of the bolalipid molecules and thus model different types of bolalipids. The model is simulated within molecular dynamics implemented in the LAMMPS open source package [[34](#) [right](#)], and simulations are visualized with OVITO [[35](#) [right](#)]. To include the implicit effect of the surrounding water and to simulate membranes at vanishing tension, we used a Langevin thermostat combined with a barostat that kept the membrane at zero pressure in the  $x$ - $y$  plane. Details on the computational model are given in SI section 1.

## Self-assembly and phase behavior

We first tested the ability of both the bilayer and bolalipid molecules to self-assemble into membranes. To do so, we placed dispersed lipids in a periodic 3D box. For both bilayer and bolalipids we found that membranes self-assembled over a wide range of lipid interaction parameters (**Fig. 1D** [right](#), SI section 2, and Movie S1). We then explored the influence of flexibility in the bolalipid molecule. At small values of the molecular rigidity  $k_{\text{bola}}$ , single lipids are flexible and can thus adopt a range of possible conformations. The different conformations can be classified by the angle  $\theta$  between the two lipid heads (**Fig. 1E** [right](#)). Two lipid conformations dominated the conformation distribution in the context of a membrane: the U-shaped conformation with both head beads on the same membrane leaflet ( $\theta \approx 0$ ) and the straight conformation with one head bead in each opposing membrane leaflet ( $\theta \approx \pi$ ) (Movie S2.b, Fig. S1, and SI section 3; for comparison, bilayer membranes in Movie S2.a and rigid bolalipid membranes in Movie S2.c). In the self-assembled bolalipid membrane, we marked bolalipids as being in the U-shape conformation if  $\theta < \pi/2$  and in the straight conformation otherwise.

To study the phase behaviour of bilayer and bolalipid membranes, we analysed the diffusion of single lipids as a function of the lipid interaction parameters  $\epsilon_p$  and  $\omega$  (SI sections 4 and 5). The diffusion constant  $D$  exhibited a discontinuity as a function of the temperature, which marks the transition as the membrane moves from the gel phase to the liquid phase. The discontinuity occurred at different values of interaction strength  $\epsilon_p$  and interaction range  $\omega$  for bilayer and bolalipid lipids, and it also depended on the values of the molecular stiffness  $k_{\text{bola}}$  (Fig. S4). In these simulations, the disintegration of the membrane defined an upper limit to the liquid phase and the transition to the gas phase. Based on this classification, we plotted the phase diagram for bilayer membranes (**Fig. 1F** [right](#) top left), fully flexible bolalipid membranes ( $k_{\text{bola}} = 0$ , **Fig. 1F** [right](#) top right), and rigid bolalipid membranes ( $k_{\text{bola}} = 5 k_B T$ , **Fig. 1F** [right](#) bottom left) as a function of the range of the hydrophobic interaction  $\omega$  and the temperature  $T_{\text{eff}}$ . Membranes made of bilayers (blue) and flexible bolalipid molecules ( $k_{\text{bola}} = 0$ , orange) behaved similarly under these conditions (**Fig. 1F** [right](#) bottom right). Strikingly, just as observed in extremophile archaea, as the molecular stiffness of bolalipids increased ( $k_{\text{bola}} = 5 k_B T$ , magenta), the liquid region was shifted toward higher temperatures and larger values of the interaction range. This is due to the fact that bolalipid molecules are able to engage in more extensive interactions with partners when in the extended conformation, which helps to stabilize the membrane at higher temperatures.

## Bolalipid conformations and mechanical properties

To explore mechanical properties of bolalipid membranes, we chose  $\omega = 1.5 \sigma$  for the remainder of the work (dashed line in **Fig. 1F** [right](#) bottom right). **Fig. 2A** [right](#) shows the phase diagram for bolalipid membranes replotted as a function of temperature and bolalipid rigidity for the chosen

interaction range. To be able to compare membranes made of bolalipids of different molecular rigidities, we needed to adjust temperature for each to reach similar fluidities, shown by dashed lines in **Fig. 2A**. We then characterised the conformations of individual bolalipids in flat membranes, measured by the fraction of bolalipids in U-shape conformation  $u_f$ . We find that for flexible bolalipids ( $k_{\text{bola}} = 0$ ), more than 50% of all lipids are in the U-shape conformation (**Fig. 2B**). This fraction decreases with increasing rigidity of bolalipid molecules and vanishes around  $k_{\text{bola}} \geq 2 k_B T$ , for which almost all bolalipids take up linear conformations.

This behaviour can be easily captured by considering bolalipids as a two state system, with straight and U-shaped conformations, as argued before (Fig. S1). We assumed that the energy of the straight conformation vanishes  $E_s = 0$  and the energy of the U-shaped conformation reads  $E_u = c_0 + c_1 k_{\text{bola}}$ , where, for simplicity, we assumed that the energy linearly depends on  $k_{\text{bola}}$  which is the only relevant energy scale in the system and the two constants  $c_0$  and  $c_1$ , which determine the conformation energy of U-shaped bolalipids. The fraction of bolalipids in U-shape conformation then follows  $u_f(k_{\text{bola}}) = 1/(1 + \exp(\beta(c_0 + c_1 k_{\text{bola}})))$ , with  $\beta = 1/(k_B T)$ , as shown by the fit in **Fig. 2B** (dashed gray line,  $R^2 = 0.99$ , see SI section 6). For the fit it appears that  $c_0 < 0$ , which implies that bolalipids in U-shape conformation are slightly favoured over straight bolalipids at  $k_{\text{bola}} = 0$  (explored in SI section 6).

## Bolalipid conformations and membrane rigidity

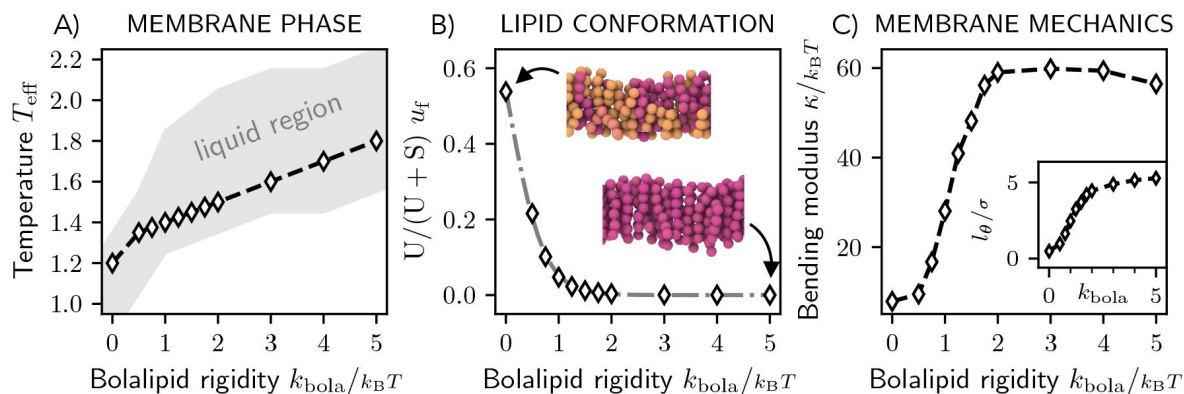
It was previously hypothesized that an increasing fraction of bolalipids in straight configuration would increase the membrane rigidity [18]. To determine the membrane rigidity using our model, we assessed the height fluctuation spectrum ( $h^2$ ) of flat membranes in a periodic box. [32, 36]. Interestingly, we found that the original theory of Helfrich [37] failed to describe the resulting height fluctuation spectrum. However, the extended theory by Hamm and Kozlov [38], which also includes the energetic cost of lipid tilt, successfully captured bolalipid fluctuations (SI section 16). In this case, the resulting height spectrum of the membrane at vanishing membrane tension is given by [36]

$$\begin{aligned} \langle |h_n|^2 \rangle &= \frac{k_B T}{L^2} \left( \frac{1}{\kappa q^4} + \frac{1}{\kappa_\theta q^2} \right) \\ &= \frac{k_B T}{L^2} \frac{1}{\kappa q^4} (1 + q^2 l_\theta^2), \end{aligned} \quad (1)$$

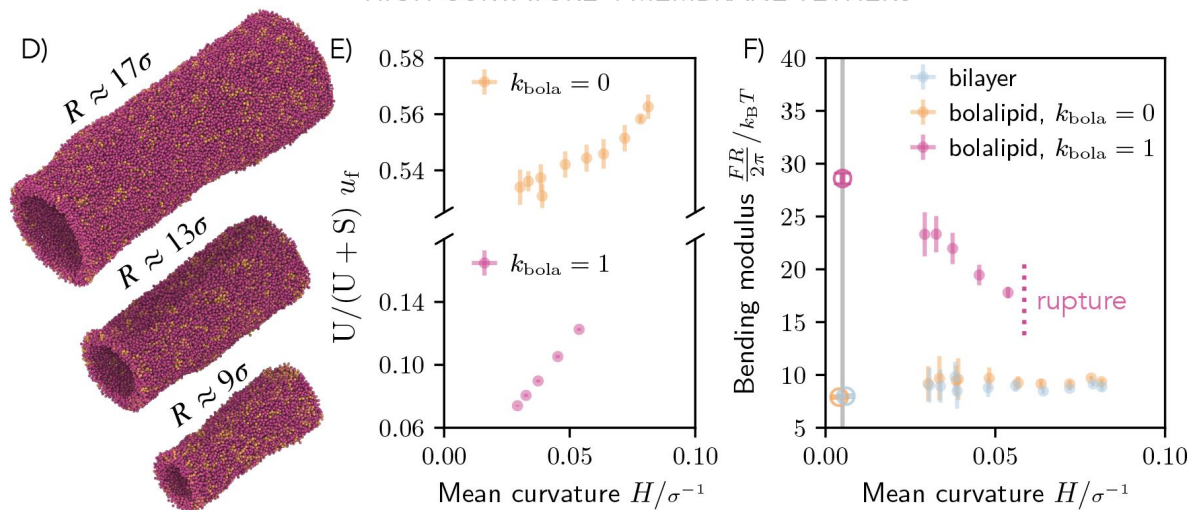
where  $q = 2\pi n/L$  is the wave number ( $n \in \mathbb{Z}$ ),  $L$  is the box size,  $\kappa$  is the bending rigidity of the membrane,  $\kappa_\theta$  is the tilt modulus and  $l_\theta = \sqrt{\kappa/\kappa_\theta}$  is a characteristic length scale related to tilt, which we call the tilt persistence length. Considering Eq. (1), the tilt term is expected to matter if the analysed inverse wave numbers become of the same order as the tilt persistence length  $l_\theta$ . The wave numbers that we analysed correspond to wavelengths that are at least twice the thickness of the membrane ( $q < 2\pi/(12\sigma) \approx 0.5 \sigma^{-1}$ ). Thus, the tilt term is expected to noticeably contribute if  $l_\theta > 2\sigma$ . For typical bilayer membranes, one finds  $\kappa_\theta = 12 k_B T \text{ nm}^{-2}$  [36] and  $\kappa = 20 k_B T$  [39], so  $l_\theta \sim 1 \text{ nm} \approx 1\sigma$  and therefore the tilt term can be neglected as practised before [32]. However, when the membrane rigidity increases as we expect for bolalipid membranes, the tilt persistence length  $l_\theta$  increases and the tilt term in Eq. (1) becomes relevant.

By fitting the height spectrum for bolalipid membranes (and bilayer membranes for comparison) (Fig. S6), we measured the bending rigidity (**Fig. 2C**) and the tilt persistence length  $l_\theta$  (**Fig. 2C** inset) as a function of  $k_{\text{bola}}$  (see SI section 7 and Movie S3). With increasing bolalipid molecular rigidity  $k_{\text{bola}}$ , the bending rigidity  $\kappa$  rose from  $8 k_B T$  and plateaued at  $60 k_B T$ , showing bolalipid membranes can be very rigid while liquid. Strikingly, the increase in membrane rigidity coincided with U-shaped bolalipids vanishing from the membrane (**Fig. 2B**), which confirmed the hypothesis that straight bolalipids render lipid membranes rigid. At the same time, the tilt persistence length  $l_\theta$  increases with bolalipid rigidity, starting near zero, crossing the  $2\sigma$  threshold at  $k_{\text{bola}} = 1 k_B T$  and plateauing at  $5\sigma$ . Since membrane bending and lipid tilting are two modes of

## LOW CURVATURE : FLAT MEMBRANES



## HIGH CURVATURE : MEMBRANE TETHERS



**Figure 2.**

### Mechanics of pure bolalipid membranes.

(A) Liquid region as a function of temperature and bolalipid rigidity for pure bolalipid membranes (gray). The dashed line shows the bolalipid membranes of approximately same fluidities. (B) Fraction of bolalipids in the U-shape conformation ( $\theta = 0$ ), fitted to  $u_f(k_{\text{bola}}) = 1/(1 + \exp(\beta(-0.16 + 3k_{\text{bola}})))$  (gray dashed line) according to a two-state model. Insets: simulation snapshots with bolalipids coloured according to their conformations. (C) Bending modulus as a function of bolalipid molecule rigidity  $k_{\text{bola}}$ . Inset: Tilt persistence length  $l_\theta$  as a function of bolalipid rigidity  $k_{\text{bola}}$ . (D) Snapshots of bolalipid membranes at the range of explored curvatures for  $k_{\text{bola}} = 1k_B T$ . (E) Fraction of bolalipid molecules in the U-shaped conformation as a function of the mean membrane curvature  $H = 1/(2R)$  for membranes made of flexible ( $k_{\text{bola}} = 0$ ) and semi-flexible ( $k_{\text{bola}} = 1k_B T$ ) bolalipid molecules. (F) Bending modulus as a function of curvature. For the flat membrane ( $H \approx 0$ ), the corresponding bending rigidity from (C) is marked by the vertical line and empty circles.

membrane deformations that compete, we conclude that bilayer and flexible bolalipids molecules form flexible membranes that prefer to bend rather than to tilt, while bolalipids in straight configuration form rigid membranes that prefer to tilt rather than to bend. Taken together, bolalipid membranes made of flexible lipid molecules are as flexible as lipid bilayers, adopting U-shaped conformations, where those made of bolalipids in straight configurations are rigid.

## Interplay between membrane curvature and rigidity

Changing membrane curvature alters the area differently in the two membrane leaflets. To adapt to the area difference, we thus expect the fraction of U-shaped bolalipids to change as the membrane curvature changes. Moreover, the results of [Fig. 2B](#) and [Fig. 2C](#) showed that the U-shaped bolalipid fraction and the membrane bending rigidity are correlated. As a result, we predict that the fraction of straight versus U-shaped bolalipids in a membrane will change in response to membrane bending, in a way that makes the bending rigidity of a bolalipid membrane curvature dependent.

To investigate the effect, we measured the bending rigidity of bolalipid membranes in cylindrical shapes as a function of their radii [\[40\]](#) (see [Fig. 2D](#), [Movie S4](#), and [SI section 8](#)). We first noticed that while membrane tubes made of bilayer and flexible bolalipids were stable up to small cylinder radii  $R$ , almost as small as the membrane thickness itself, we found that membrane made from stiffer bolalipids ( $k_{\text{bola}} = 1k_{\text{B}}T$ ) ruptured well before. Strikingly, while for stiffer bolalipids ( $k_{\text{bola}} = 1k_{\text{B}}T$ ) the U-shaped bolalipid fraction increased strongly over a short range of the mean membrane curvature ( $H = 1/(2R)$ ), we only found a small change in the U-shaped bolalipid fraction of flexible bolalipid membranes ( $k_{\text{bola}} = 0$ ) ([Fig. 2E](#)). Consequently, since the fraction of U-shaped bolalipid molecules controls the bending rigidity of bolalipid membranes ([Figs. 2B](#) and [2C](#)), we found that there is a strong dependency of the bending rigidity  $\kappa$  on the membrane mean curvature of stiffer bolalipids ( $k_{\text{bola}} = 1k_{\text{B}}T$ ) ([Fig. 2F](#)).

In an elastic material, the strain modulus holds constant and deformation is reversible. For bolalipid membranes at  $k_{\text{bola}} = 1k_{\text{B}}T$ , however, the bending modulus decreases when deformation increases, rendering bolalipid membranes hypoelastic. Fortunately, this dependency on curvature does not invalidate our fluctuation results, where the curvature is small enough that its effect on the bending modulus is negligible ([SI section 15](#)). In contrast, we did not find that  $\kappa$  was curvature-dependent for bilayer or flexible bolalipid membranes ( $k_{\text{bola}} = 0$ ). Taken together, the rigidity of bolalipid membranes is not only controlled by the molecular stiffness of their lipid constituents but also by the emerging geometry of the ensemble of lipids. Since membrane geometry and thus membrane rigidity will change upon membrane deformations this gives rise to hypoelastic material properties.

## Gaussian rigidity of bolalipid membranes

Another important material parameter is the Gaussian bending modulus  $\bar{\kappa}$ , which characterizes the reshaping behaviour of fluid lipid membranes under topological changes [\[39\]](#).  $\bar{\kappa}$  is notoriously difficult to measure since it only becomes detectable when the membrane changes its topological state. Continuum membrane theory, combining shape stability arguments and elasticity, predicts  $-\bar{\kappa}/\kappa \in [-0.5, -1]$  [\[39\]](#), where the former value is expected for incompressible membranes. Indeed, most of the numbers we know for the ratio of the two bending rigidities, many of which were deduced from simulations, lie within this range [\[41\]](#). Using the same method as developed by Hu et al. [\[41\]](#), we determined  $\bar{\kappa}$  by measuring the closing efficiency of membrane patches into a sphere (see [SI section 9](#)). At  $T_{\text{eff}} = 1.2$ , we obtained  $\bar{\kappa} = -4.30 \pm 0.22 k_{\text{B}}T$  and thus a ratio of  $-\bar{\kappa}/\kappa = 0.89 \pm 0.04$  for bilayer membranes, similar to what has been reported previously [\[41\]](#). For flexible bolalipid membranes, we got a slightly smaller value for  $\bar{\kappa} = -5.04 \pm 0.37 k_{\text{B}}T$ . Due to the larger bending modulus, however, flexible bolalipid membranes show a significantly smaller ratio  $-\bar{\kappa}/\kappa = 0.64 \pm 0.04$  ( $k_{\text{bola}} = 0$ ). At larger temperature ( $T_{\text{eff}} = 1.3$ ),

the ratio can be even smaller  $-\bar{\kappa}/\kappa = 0.45 \pm 0.07$  (see SI section 9). The result shows that in addition to the differences in bending rigidities, the ratio of the two bending moduli differs strongly in bilayer and bolalipid membranes.

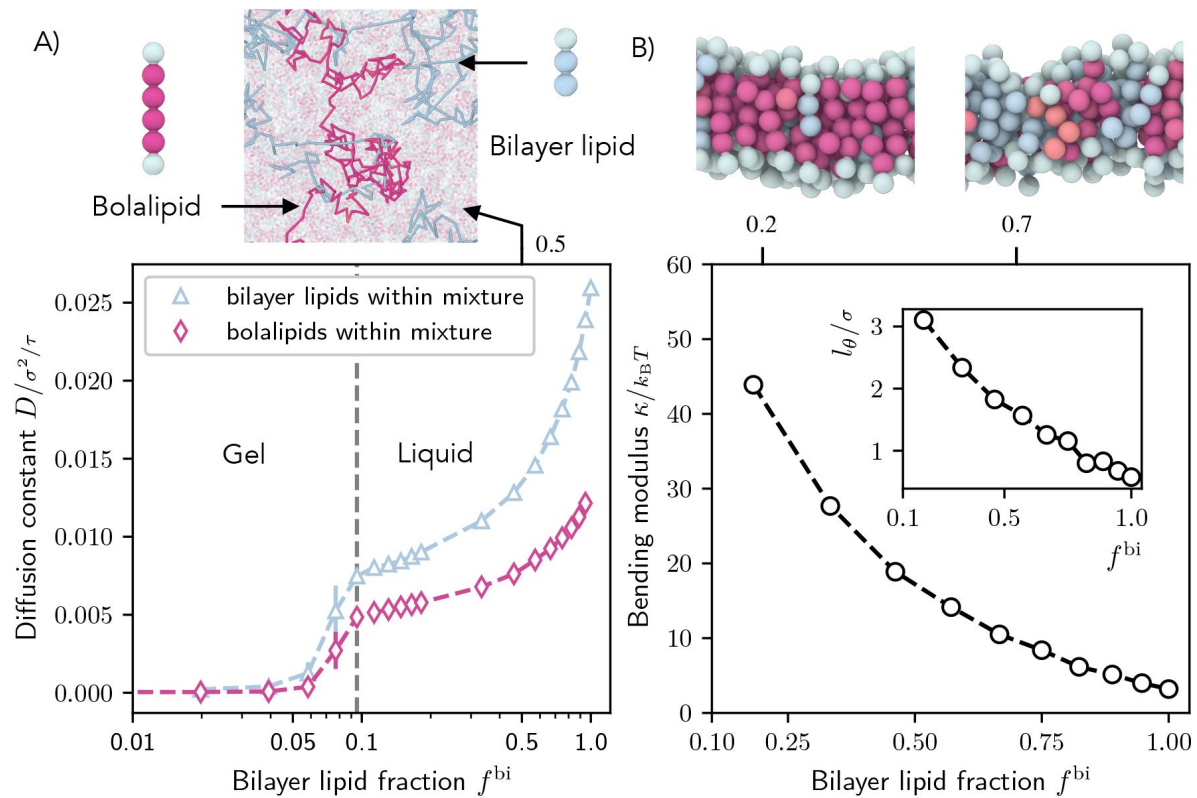
## Archaeal membranes made of mixtures of bolalipids and bilayer-forming lipids

Archaeal membranes contain varying amounts of bilayer lipids [5, 8]. The exact bolalipid/bilayer fraction depends on the growth temperature, with higher levels of bolalipids with increasing temperature [9], and higher fraction of cyclopentane rings in the tails [33]. In order to investigate the effect of different lipid contents on membrane mechanical properties, we wanted to model the archaeal membrane by mixing bilayer lipids into bolalipid membranes. Since in our model, the liquid regions of rigid bolalipid membranes and bilayer membranes do not overlap (Fig. 1F bottom right), we picked the temperature  $T_{\text{eff}} = 1.3$  to minimize fluidity mismatch and we set the molecular rigidity  $k_{\text{bola}} = 2 k_{\text{B}}T$  to limit U-shaped bolalipids (Fig. 2B). We then measured the diffusion constant  $D$  as a function of the fraction of bilayer lipids  $f^{\text{bi}}$  (see SI section 5). Interestingly, we found that mixing only 10% bilayer lipids into the bolalipid membrane in gel state is enough to fluidize the membrane (Fig. 3A). We then measured the bending rigidity and the tilt persistence length  $l_{\theta}$  of flat mixture membranes by analysing the fluctuation spectrum. Both the bending rigidity  $\kappa$  (Fig. 3B) and the tilt persistence length  $l_{\theta}$  decreased non-linearly with the bilayer lipid fraction  $f^{\text{bi}}$  (Fig. 3B inset). Taken together, the bolalipid membrane can be substantially softened either through bolalipids acquiring the U-shaped conformation or through addition of bilayer-forming lipids.

## Curving bolalipid membranes

To investigate the response of bolalipid membranes to large membrane curvature and topology changes like those induced upon vesicle budding, which regularly occurs in archaea, we simulated membrane wrapping of an adhesive cargo bead (Fig. 4A, Movies S5.a to S5.c, and SI section 10). Importantly, this provided us with a method to study how lipid organization is affected by externally imposed membrane curvature and mechanics. We first simulated membrane wrapping at different adhesion energies  $\epsilon_{\text{mc}}$  between lipid head beads and the cargo until we observed that the membrane wrapped the cargo completely (including membrane fission). Then the minimum adhesion energy, for which a membrane bud completely enveloped the cargo bead, is the onset adhesion energy  $\epsilon_{\text{mc}}^*$  (Fig. 4A), which we measure as a function of the bolalipid stiffness  $k_{\text{bola}}$  (Fig. 4B). For small molecular stiffness  $k_{\text{bola}}$ ,  $\epsilon_{\text{mc}}^*$  first increases linearly with  $k_{\text{bola}}$  before it saturates around  $k_{\text{bola}} = 3 k_{\text{B}}T$ . We expect that the onset energy is proportional to the membrane bending rigidity  $\epsilon_{\text{mc}}^* \propto \kappa$ , because the bending energy to wrap a spherical particle is size-invariant [39, 42]. When we increased the bending rigidity, through increasing stiffness of bolalipid molecule  $k_{\text{bola}}$ ,  $\epsilon_{\text{mc}}^*$  increased by a factor of 3 (Fig. 4B), suggesting that also  $\kappa$  increased by a factor of 3. However, from directly measuring the membrane rigidity from the fluctuation spectrum (Fig. 2C), we saw that  $\kappa$  increased by a factor of 10. To reconcile these seemingly conflicting observations we reason that the bending rigidity  $\kappa$ , similar to Fig. 2F, is not constant but softens in the range  $k_{\text{bola}} \in [0, 2 k_{\text{B}}T]$ , upon increasing membrane curvature. This is due to the dynamic change in the ratio between bolalipids in straight and U-shaped conformation. Hence, bolalipid membranes show marked hypoelastic behaviour as they soften during reshaping.

Through analysing the bolalipid conformations, we found that the membrane was able to curve by increasing the fraction of U-shaped bolalipids in the outer layer of the deformation (Fig. 4C and SI section 12). To a lesser degree, we also observed this effect on the inner membrane neck (Fig. S14). Remarkably, though, even when lipids are so stiff that there are no more U-shaped bolalipids in the flat mother membrane ( $k_{\text{bola}} = 2 k_{\text{B}}T$ ), the outer layer of the a curved membrane retained a non-negligible fraction of  $\approx 10\%$  U-shaped bolalipids, which in turn decreases  $\kappa$  and softens the membrane. However, the softening effect on the membrane, indicated through a constant onset

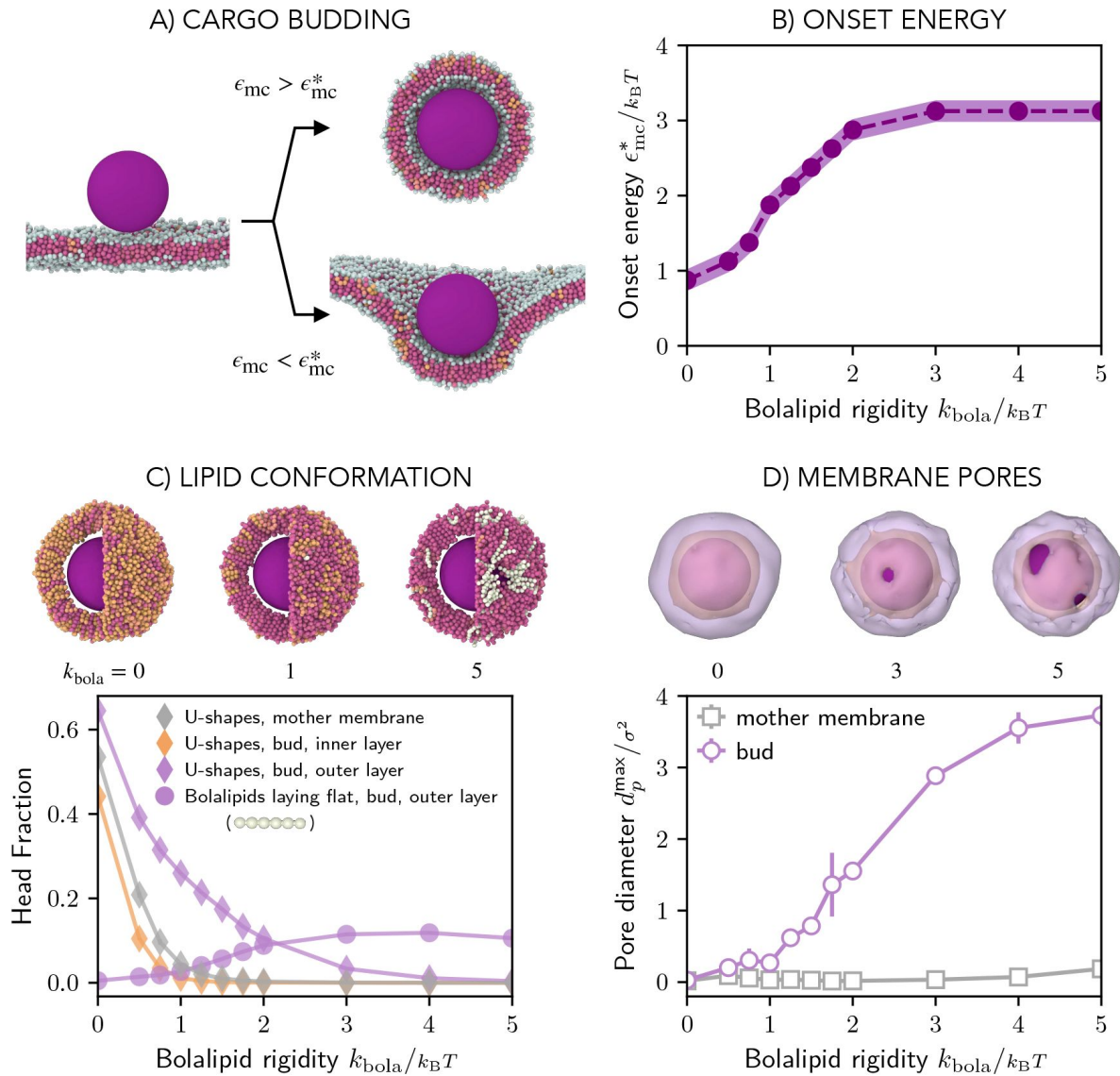


**Figure 3.**

#### Fluidity and rigidity of mixed bilayer/bolalipid membranes.

(A) Single lipid diffusion constant for each species as a function of bilayer fraction  $f^{bi}$  ( $k_{bola} = 2k_B T$ ,  $T_{eff} = 1.3$ ). For  $f^{bi} \geq 0.1$ , the resulting mixture becomes liquid. Top: Diffusion trajectories of a bolalipid (crimson) and a bilayer lipid (blue) in a mixture membrane at  $f^{bi} = 0.5$ . (B) Bending rigidity  $\kappa$  and (Inset) tilt persistence length  $l_\theta$  as a function of the fraction of bilayer molecules  $f^{bi}$ . Top: Snapshots show bilayer lipids (blue) in mixed membranes at two different values of  $f^{bi}$ .

## PURE BOLALIPID MEMBRANES



**Figure 4.**

### Reshaping of pure bolalipid membranes.

(A) Simulation snapshots of the membrane wrapping a cargo bead adhering to it. Above the onset adhesion energy  $\epsilon_{mc}^*$ , the cargo is fully wrapped by the membrane and buds off the mother membrane. (B) Onset energy  $\epsilon_{mc}^*$  as function of the bolalipid molecule rigidity  $k_{bola}$  (for the parameters defined by the line in [Fig. 2A](#)). (C) Bottom: Fraction of bolalipids in the U-shape conformation  $u_f$  in the outer and inner layers of the membrane bud, and in the flat mother membrane, as function of the bolalipid molecule rigidity  $k_{bola}$ . Top: Snapshots and cross-sections of the membrane around the cargo bud. At high bolalipid rigidity the pores form around the cargo, and are lined with bolalipid molecules lying flat around the pore in a straight conformation, with both heads in the outer layer (coloured in white). The rest of bolalipids coloured according to their head-to-head angle as before. (D) Bottom: Average diameter of transient pores in the membrane bud and the mother membrane as function of the bolalipid molecule rigidity  $k_{bola}$ . Pores are defined as membrane openings through which a sphere of diameter  $1\sigma$  can cross. Top: Snapshots of the membrane surface with outer and inner leaflet surface coloured in purple and orange, respectively, intersecting at the rim of the pore (gray).

energy for  $k_{\text{bola}} \geq 3 k_B T$  (Fig. 4B), persists even for those very stiff bolalipids. Since for stiff membranes, practically all U-shaped bolalipids are gone (Fig. 4C), this suggested that an additional membrane-curving mechanism must be involved.

Looking more closely, at high molecular rigidity ( $k_{\text{bola}} \geq 2k_B T$ ) we observed the formation of multiple pores on the membrane bud, which we quantified by measuring the time-averaged maximum pore diameter (Fig. 4D, see SI section 12 and Movie S6). While large pores were not observed in the flat membrane, the diameter of membrane pores around the cargo was found to grow with the increase in bolalipid stiffness. We reasoned that pores form when the energetic cost required to change the bolalipid conformation to release bending stress is larger than the energetic cost of opening a lipid edge surrounding the pore. Hence, for relatively flexible bolalipids, U-shaped bolalipids provide the necessary area difference between the outer and inner layer of the membrane bud and thereby soften the membrane. For stiff bolalipid molecules, however, membrane pores start to form to enable membrane curvature as U-shaped bolalipids become prohibited. Both mechanisms help to explain the discrepancy between  $\epsilon_{\text{mc}}^*$  and the bending modulus  $\kappa$  obtained by studying membrane fluctuations (Fig. 2C).

## Curving archaeal membranes

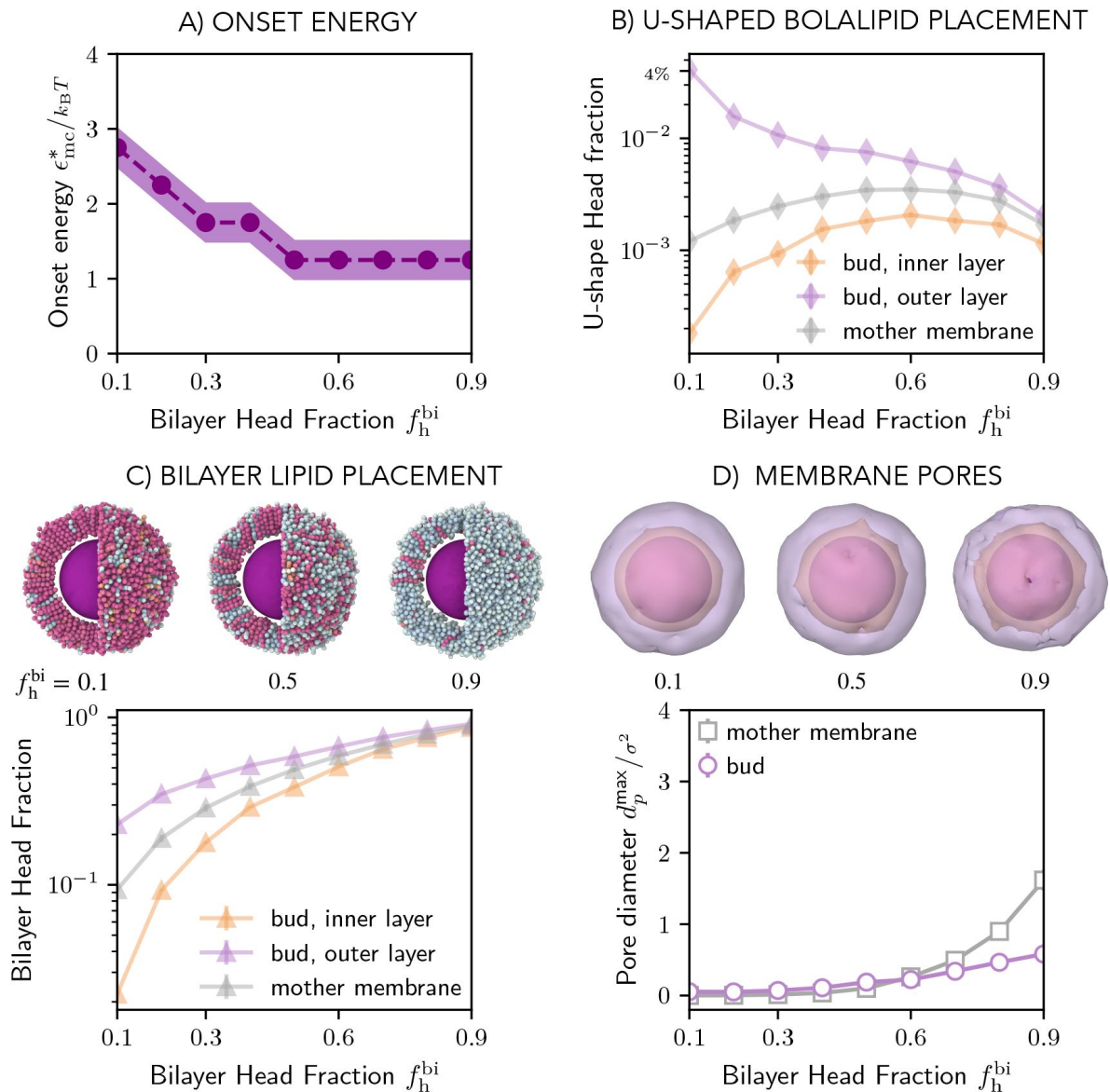
Having shown that bolalipid membranes can effectively soften also by including some amount of bilayer-forming lipid molecules, we next measured the onset energy  $\epsilon_{\text{mc}}^*$  for cargo budding in the membranes formed by mixtures of bolalipids and bilayer-forming lipids, as a function of bilayer lipid head fraction  $f_{\text{h}}^{\text{bi}}$  (Fig. 5A and Movie S5.d). We found that the onset energy sharply decreases with increasing amount of bilayer forming lipids, and plateaus for 50% bilayer head fraction, where it acquires similar values as in the case of fully-flexible bolalipids (Fig. 4B). For small bilayer fractions, U-shaped bolalipids localize almost exclusively on the outer layer of the bud (Fig. 5B). As the bilayer fraction increases, there is a steady reduction in the percentage of U-shaped bolalipids in the outer layer in favour of bilayer lipids that take their role in supporting membrane curvature, with U-shaped bolalipids completely vanishing at high fractions of bilayer-forming lipids. The fraction of bilayer lipid head beads initially shows an asymmetry between the preferred outer layer and the penalized inner layer around the cargo (Fig. 5C), but eventually approaches  $f_{\text{h}}^{\text{bi}}$  in both layers. Taken together, as the bilayer lipid fraction increases, the role of U-shaped bolalipids in making up the asymmetry between the outer and inner layer, is taken over by bilayer lipids. Curiously, the addition of bilayer lipids promotes the formation of U-shaped lipids, both in the flat membrane and even in the inner layer around the bud. This is likely to be explained by the fact that when more bilayer lipids are incorporated, the membrane is less densely packed (Fig. S12B) and thus U-shaped bolalipids are promoted.

Importantly, we observed nearly no pores in the membrane bud in mixed membranes, even when we only had a very small fraction of bilayer-forming lipids (Fig. 5D). Only as the bilayer fraction increased, did we observe the formation of very small pores in the bud. For the flat mother membrane, however, membrane pores started to form with increasing values of  $f_{\text{h}}^{\text{bi}}$ . They acquired sizes similar to those obtained around the bud in pure bolalipid membranes (Fig. 4D). The pore formation in the flat mother membrane is likely promoted because the membrane becomes destabilized by the increasing proportion of bilayer lipids which are close to the gas phase. Taken together, bolalipids can bud porelessly when bilayer-forming lipids, which cause membrane softening, are included.

## Discussion

All biological cells are enclosed in fluid lipid membranes. These must be both sturdy and flexible to contain the cellular interior and allow membrane reshaping during division and vesicle formation. Across nature, mono- and bilayer membranes, composed of bolalipids and bilayer

## MIXTURE OF BOLALIPIDS AND BILAYER-FORMING LIPIDS



**Figure 5.**

### Curving of the mixed membranes, made of bilayer and bolalipid molecules.

(A) Onset energy required to form the membrane bud,  $\epsilon_{mc}^*$ , as function of bilayer head fraction  $f_h^{bi}$  (for the parameters defined in **Fig. 3**). (B and C) Fraction of U-shaped bolalipid molecules  $u_f$  (B) and bilayer molecules (C) in the outer and inner layers of the membrane bud and in the flat mother membrane as a function of the bilayer head fraction  $f_h^{bi}$ . Top panels show the respective snapshots of membrane surface around cargo, where bilayer lipids are shown in light blue as in **Fig. 4**. (D) Average diameter of transient pores in the membrane bud and the mother membrane as function of bilayer head fraction  $f_h^{bi}$  and respective snapshots of membrane leaflet surfaces surrounding the bud (top panel).

lipids, have evolved to fulfil these partially conflicting requirements. In this work, we have developed the first minimal model that allows us to transparently compare the distinct behaviour of bilayer and bolalipid membranes. In our model, bolalipids are formed by joining together two bilayer lipids, with an adjustable molecular stiffness at the hinge point. Using our model we find striking differences between bilayer and bolalipid membranes in terms of thermal and mechanical stability, bending rigidity and presence of hypoelasticity. Our results highlight how bolalipid membrane behave effectively as two-component systems, exchanging bolalipids between different conformations and thereby allowing the material properties to be adapted to support membrane reshaping.

While flexible bolalipid membranes are liquid under the same conditions as bilayer membranes, we found that stiff bolalipids form membranes that operate in the liquid regime at higher temperatures. These results agree well with previous molecular dynamics simulations that suggested that bolalipid membranes are more ordered and have a reduced diffusivity compared to bilayer membranes [24, 29]. In our simulations, this is due to the fact that completely flexible bolalipids molecules adopt both straight (transmembrane) as well as the U-shaped (loop) conformation with approximately the same frequency. In contrast, stiff bolalipids typically only take on the straight conformation when assembled in a membrane. These results agree with the previous coarse-grained molecular dynamics simulations using the MAR-TINI force field which showed that the ratio of straight to U-shaped bolalipids increased upon stiffening the linker between the lipid tails [29].

While previous coarse-grained simulations predicted that bolalipids spontaneously transition between the straight and U-shaped conformations [29], how this happens in archaeal membranes and whether membrane proteins are involved in this conformational transition needs to be clarified in the future. Experimental studies suggest that archaeal membranes contain flippases and scramblases for the transitioning of bilayer lipids between membrane leaflets [43, 44], raising the possibility that similar proteins could also facilitate conformational transitions in bolalipids. In addition, it has been suggested that the viral fusion protein hemagglutinin could cause a transition from straight to U-shaped bolalipid conformation during the fusion of bolalipid vesicles with influenza viruses [17]. However, future investigation is required.

When we determined the bending rigidity of bolalipid membranes by measuring their response to thermal fluctuations, we found that membranes made from flexible bolalipids are only slightly more rigid than bilayer membranes. This result is consistent with previous atomistic simulations, which showed that the membrane rigidity was similar for membranes composed of bilayer lipids and flexible synthetic bolalipids [45]. Moreover, the result is consistent with a continuum theory which predicted that the rigidity of membranes formed of triblock copolymers is 20% larger than that of diblock copolymers [46]. However, bolalipids in extremophilic archaea are not predicted to be fully flexible as they are expected to pack tighter due to a large number of cyclopentane rings in the lipid tails [7, 33]. Indeed, we found that membranes made of stiff bolalipid molecules can exhibit stiffness that is more than an order of magnitude larger than that of bilayer lipids at the same membrane fluidity.

It is striking that membranes made from stiffer bolalipids showed a curvature-dependent bending modulus, which is a clear signature that bolalipid membranes exhibit hypoelastic behaviour during membrane reshaping. Another marked difference between bilayer and flexible bolalipid membranes is the ratio of the Gaussian rigidity to the bending modulus. Instead of being around  $-1$  as for bilayer membranes [41], it is around  $-1/2$  and therefore only half of that of bilayer lipids. It is not obvious how the Gaussian bending modulus would behave upon increasing bolalipid stiffness ( $k_{\text{bola}} > 0$ ), or how to measure it due to the coupling between curvature and rigidity in bolalipid membranes. Membrane remodelling, such as the fission of one spherical vesicle into two, increases the bending energy by  $8\pi\kappa$  but decreases the energy related to the Gaussian modulus by  $-4\pi|\bar{\kappa}|$  [39], giving rise to a fission energy barrier of  $\Delta E_{\text{fission}} = 4\pi\kappa(2 - |\bar{\kappa}|/\kappa)$ .

Our results indicated that while in bolalipid membranes  $\kappa$  is larger,  $|\bar{\kappa}|/\kappa$  is smaller compared to bilayer membranes. Our results thus predict a larger energy barrier for membrane fission  $\Delta E_{\text{fission}}$  in bolalipid membranes compared to bilayer membranes. It is tempting to speculate that this is one of the reasons why eukaryotes use bilayer membranes, enabling dynamic membrane remodelling and trafficking.

It is interesting to draw a parallel between monolayer membranes made of stiff bolalipid molecules and macroscopic membranes composed of rigid straight colloidal particles, which are geometrically similar, but living at different scales [47]. It has been found that colloidal membranes at these macroscopic scales follow the standard Helfrich theory for bilayer membranes [48], with rigidity that is three orders of magnitude higher than those of lipid bilayers [49]. In this case the tilt modulus was not pertinent, likely due to macroscopic system sizes. Similarly, we expect that the bending rigidity can be determined from membrane fluctuations independently of the tilt modulus for bolalipid membranes if they are prepared at similar relative sizes. Beyond quantitative differences, the comparison shows that monolayer membranes follow the same physics across many orders of magnitudes. However, when considering subcellular scales and the formation of high curvature, as in vesicle budding, the tilt modulus and thus the substructure of the membrane is expected to matter.

Our model makes a number of predictions that could be tested by experiment either in cells or in vitro. First, it predicts that a small increase in the fraction of archaeal bilayer lipids should be sufficient to soften a bolalipid-rich membrane. While this could be tested in the future, so far only very few studies have yet reported experimental analysis of archaeal membrane mixtures [18, 50]. Second, we observed that membranes with moderate bolalipid molecular rigidity  $k_{\text{bola}}$  exhibit curvature-dependent bending rigidity. To experimentally verify this, one could extrude membrane tethers from cells while controlling for membrane tension. Finally, to get to the core mechanism underlying our findings, it will be important to develop experimental methods that will allow the fraction of U-shaped bolalipid conformers per leaflet to be imaged and measured.

We found that membranes formed of a mixture of bilayer and bolalipids, similar to archaeal membranes, function as a composite liquid membrane that softens when adding bilayer lipids. However, while in our simulations the bending rigidity monotonically decreases with bilayer fraction, previous experiments of mixture membranes of bilayer and bolalipids with cyclopentane rings suggested that the bending rigidity non-monotonically depends on the fraction of the membrane made up of bilayer lipids [18]. It remains to be determined whether the result is due to the specific lipids used, the resulting mismatch in lengths between two stacked bilayer lipids and a straight bolalipid, the experimental conditions or to non-linear effects such as the formation of lipid domains that soften the membrane with increasing bolalipid content. The same experiments reported that membranes consisting solely of bolalipids are more rigid than bilayer membranes and non-fluctuating, which is in agreement with our high bending modulus simulation results for nearpure bolalipid mixture membranes.

To investigate how bolalipid membranes respond to changing membrane curvature, we performed simulations in which small cargo particles budded from flat membranes. We found that by enforcing curvature on bolalipid membranes, the fraction of U-shaped bolalipids increased around the cargo bud, especially in the outer membrane layer and hence softened the membrane. As another mechanism to release curvature stress we observed the formation of membrane pores, which could be mended by adding small amounts of bilayer lipids, similar to the mixture membranes that are found in archaea [8]. Our results suggest that enforcing membrane bending can soften bolalipid membranes locally by increasing the number of U-shapes, rendering the membrane a mechanically switchable material where large curvature decreases stiffness.

Taken together, our results show how membranes which are mixtures of bilayer and bolalipids maintain cell integrity at high temperatures, while also undergoing leak free membrane bending. This suggest that archaeal membranes can balance opposing needs when adapting to extreme environmental conditions. Beyond understanding membrane properties and reshaping across the tree of life, these results pave the way for synthetic bolalipid membranes and bolalipid membrane containers with new and exciting material properties.

## Data availability

The simulation input files and codes are freely available at [51 [↗](#)].

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## Additional information

### Author contributions

MA performed computer simulations and analysis and wrote the manuscript. FF performed patch closing simulations, contributed to computer simulations, performed theoretical analysis, and wrote the manuscript. XJ contributed to budding simulations and the manuscript. BB co-supervised the project and edited the manuscript. AŠ conceived the study, guided the project, and wrote the manuscript.

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## Additional files

**Supplemental Information** [↗](#)

**MovieS1** [↗](#)

**MovieS2a** [↗](#)

**MovieS2b** [↗](#)

**MovieS2c** [↗](#)

**MovieS3** [↗](#)

**MovieS4** [↗](#)

**MovieS5a** [↗](#)

**MovieS5b** [↗](#)

**MovieS5c** [↗](#)

**MovieS5d** [↗](#)

**MovieS6** [↗](#)

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## Reviewer #2 (Public review):

Summary:

The authors aimed to understand the biophysical properties of archeal membranes made of bolalipids. Bacterial and eukaryotic membranes are made of lipids that self-assemble into bilayers. Archaea, instead, use bolalipids, lipids that have two headgroups and can span the entire bilayer. The authors wanted to determine if the unique characteristics of archaea, which are often extremophiles, are in part due to the fact that their membranes contain bolalipids.

The authors develop a minimal computational model to compare the biophysics of bilayers made of lipids, bolalipids, and mixtures of the two. Their model enables them to determine essential parameters such as bilayer phase diagrams, mechanical moduli, and the bilayer behavior upon cargo inclusion and remodeling.

The author demonstrates that bolalipid bilayers behave as binary mixtures, containing bolalipids organized either in a straight conformation, spanning the entire bilayer, or in a u-shaped one, confined to a single leaflet. This dynamic mixture allows bolalipid bilayers to be very sturdy but also provides remodeling. However, remodeling is energetically more expensive than with standard lipids. The authors speculate that this might be why lipids were more abundant in the evolutionary process.

Strengths:

This is a wonderful paper, a very fine piece of scholarship. It is interesting from the point of view of biology, biophysics, and material science. The authors mastered the modeling and analysis of these complex systems. The evidence for their findings is really strong and complete. The paper is written superbly, the language is precise and the reading experience very pleasant. The plots are very well-thought.

Weaknesses:

None. The authors have addressed all the potential weaknesses that were raised by the reviewers.

<https://doi.org/10.7554/eLife.105432.2.sa2>

## Reviewer #3 (Public review):

Summary:

The authors have studied the mechanics of bolalipid and archaeal mixed-lipid membranes via comprehensive molecular dynamics simulations. The Cooke-Deserno 3-bead-per-lipid model is extended to bolalipids with 6 bead. Phase diagrams, bending rigidity, mechanical stability of curved membranes, and cargo uptake are studied. Effects such as formation of U-shaped bolalipids, pore formation in highly curved regions, and changes in membrane rigidity are studied and discussed. The main aim has been to show how the mixture of bolalipids and regular bilayer lipids in archaeal membrane models enhances the fluidity and stability of these membranes.

The authors have presented a wide range of simulation results for different membrane conditions and conformations. Analyses and findings are presented clearly and concisely. Figures, supplementary information and movies are of very high quality and very well present what has been studied. The manuscript is well written and is easy to follow.

The authors have provided detailed response to the points I raised on the first version and have revised their manuscript accordingly. Hence, I only mention what, in my opinion, still deserves to be noted.

Comments:

I previously raised an issue with respect to the resort to the Hamm-Kozlov model for fitting the power spectrum of membrane undulations. The authors provided very nice arguments against my concerns. For the sake of completeness, I include a simple scenario, which will better highlight the issue:

The tilt contribution to the Helfrich Hamiltonian can be written as a quadratic term  $\frac{1}{2} k_t |\mathbf{T}|^2$ , where  $\mathbf{T}$  is a tilt vector field. This field is written as the difference between the surface normal and the director field aligned with the lipid orientations. In the small deviation Monge description with  $z=h(x, y)$  as the height function, the surface normal has the form  $\mathbf{N}=(-dh/dx, -dh/dy, 1)$ . Now assume the director field,  $\mathbf{n} = (b_x, b_y, 1)$  with small  $b_x$  and  $b_y$  components. The tilt contribution to the energy thus reads as  $\frac{1}{2} k_t (\mathbf{N} - \mathbf{n})^2 \approx \frac{1}{2} k_t [|\text{grad } h|^2 + 2 \mathbf{b} \cdot \text{grad } h]$ . The first term,  $\frac{1}{2} k_t |\text{grad } h|^2$ , is indeed similar to a surface tension term,  $\sigma |\text{grad } h|^2$  that you get from the  $(1 + \frac{1}{2} |\text{grad } h|^2)$  approximation to the area element. Therefore, if you only look at height fluctuations, while your membrane actually has some surface tension, it will make distinguishing the tilt contributions to the fluctuations in the linear Monge gauge impossible.

However, considering that the authors have made sure that the membrane is indeed tensionless, this argument is settled.

I had also raised an issue about the correct NpT sampling in the simulations, and I'm glad that the authors also set up more rigorously thermostatted/barostatted simulations to check the validity of their findings.

Also, from the SI, I previously noted that the authors had neglected the longest wavelength mode because it was not equilibrated. This was an important problem and the authors looked into it and ran more simulations that were better equilibrated.

The analysis of energy of U-shaped lipids with the linear model  $E=c_0 + c_1 * k_{\text{bola}}$  is indeed very interesting. I am glad that the authors have expanded this analysis and included mean energy measurements.

<https://doi.org/10.7554/eLife.105432.2.sa1>

**Author response:**

The following is the authors' response to the original reviews

**Public Reviews:****Reviewer #1 (Public review):***Summary:*

*Amaral et al. presents a study investigating the mesoscale modelling and dynamics of bolalipids.*

*Strengths:*

*The figures in this paper are exceptional. Both those to outline and introduce the lipid types, but also the quality and resolution of the plots. The data held within also appears to be outstanding and of significant (hopefully) general interest.*

We thank the reviewer for their kind words and the appreciation of our work.

*Weaknesses:*

*In the introduction, I would like to have read more specifics on the biological role of bolalipids. Archaea are mentioned, but this kingdom is huge - there must be specific species that can be discussed where bolalipids are integral to archaeal life. The authors should go beyond 'extremophiles'. In short, they should unpack why the general audience should be interested in these lipids, within a subset of organisms that are often forgotten about.*

Following the reviewer's advice we have revised the introduction of the manuscript, in which we now discuss specific species (*Sulfolobus acidocaldarius* and *Thermococcus kodakarensis*) and how in these species bolalipids are integral to archaeal life. We explain that the ratio between bilayer and bolalipids, and the number of cyclopentane rings contained within bolalipids can change to adapt to the environment. The revised parts of the introduction read (p.1 ):

"Like for bacteria and eukaryotes, archaea must keep their lipid membranes in a fluid state (homeoviscous adaptation). This is important even under extreme environmental conditions, such as hot and cold temperatures, or high and low pH values [7]. Because of this, many archaea adapt to changes in their environment by tuning the lipid composition of their membranes: altering the ratio between bola- and bilayer lipids in their membranes [8, 9] and/or by changing the number of cyclopentane rings in their lipid tails, which are believed to make lipid molecules more rigid [5]. For example, *Thermococcus kodakarensis* increases its tetraether bolalipid ratio from around 50% to over 80% when the temperature of the environment increases from 60 to 85 C [10]. Along the same lines, the cell membrane of *Sulfolobus acidocaldarius*, can contain over 90 % of bolalipids with up to 8 cyclopentane rings at 70 C and pH 2.5 [5, 11]. It is worth mentioning that in exceptional cases bacteria also synthesise bolalipids in response to high temperatures [12], highlighting that the study of bolalipid membranes is relevant not only for archaeal biology but also from a general membrane biophysics perspective."

**Reviewer #2 (Public review):***Summary:*

*The authors aimed to understand the biophysical properties of archeal membranes made of bolalipids. Bacterial and eukaryotic membranes are made of lipids that self-assemble into bilayers. Archea, instead, use bolalipids, lipids that have two headgroups and can span the entire bilayer. The authors wanted to determine if the unique characteristics of archaea, which are often extremophiles, are in part due to the fact that their membranes contain bolalipids.*

*The authors develop a minimal computational model to compare the biophysics of bilayers made of lipids, bolalipids, and mixtures of the two. Their model enables them to determine essential parameters such as bilayer phase diagrams, mechanical moduli, and the bilayer behaviour upon cargo inclusion and remodelling.*

*The author demonstrates that bolalipid bilayers behave as binary mixtures, containing bolalipids organized either in a straight conformation, spanning the entire bilayer, or in a u-shaped one, confined to a single leaflet. This dynamic mixture allows bolalipid bilayers to be very sturdy but also provides remodelling. However, remodelling is energetically more expensive than with standard lipids. The authors speculate that this might be why lipids were more abundant in the evolutionary process. Strengths:*

*This is a wonderful paper, a very fine piece of scholarship. It is interesting from the point of view of biology, biophysics, and material science. The authors mastered the modelling and analysis of these complex systems. The evidence for their findings is really strong and complete. The paper is written superbly, the language is precise and the reading experience is very pleasant. The plots are very well-thought-out.*

*Weaknesses:*

*I would not talk about weaknesses, because this is really a nice paper. If I really had to find one, I would have liked to see some clear predictions of the model expressed in such a way that experimentalists could design validation experiments.*

We thank the reviewer for their very kind assessment. We incorporated their recommendations regarding experimental validation in the discussion section, as follows (p.14):

“Our model makes a number of predictions that could be tested by experiment either in cells or in vitro. First, it predicts that a small increase in the fraction of archaeal bilayer lipids should be sufficient to soften a bolalipid-rich membrane. While this could be tested in the future, so far only very few studies have yet reported experimental analysis of archaeal membrane mixtures [18, 50]. Second, we observed that membranes with moderate bolalipid molecular rigidity  $k_{\text{bola}}$  exhibit curvature-dependent bending rigidity. To experimentally verify this, one could extrude membrane tethers from cells while controlling for membrane tension. Finally, to get to the core mechanism underlying our findings, it will be important to develop experimental methods that will allow the fraction of U-shaped bolalipid conformers per leaflet to be imaged and measured.”

### **Reviewer #3 (Public review):**

*Summary:*

*The authors have studied the mechanics of bolalipid and archaeal mixed-lipid membranes via comprehensive molecular dynamics simulations. The Cooke-Deserno 3-bead-per-lipid model is extended to bolalipids with 6 beads. Phase diagrams, bending rigidity, mechanical stability of curved membranes, and cargo uptake are studied. Effects such as the formation of U-shaped bolalipids, pore formation in highly curved regions, and changes in membrane rigidity are studied and discussed. The main aim has been to*

*show how the mixture of bolalipids and regular bilayer lipids in archaeal membrane models enhances the fluidity and stability of these membranes.*

*Strengths:*

*The authors have presented a wide range of simulation results for different membrane conditions and conformations. For the most part, the analyses and their results are presented clearly and concisely. Figures, supplementary information, and movies very well present what has been studied. The manuscript is well-written and is easy to follow.*

We thank the reviewer for the detailed assessment of our work and their constructive feedback.

*Major issues*

*R3.Q1: The Cooke-Deserno model, while very powerful for biophysical analysis of membranes at the mesoscale, is very much void of chemical information. It is parametrized such that it is good in producing fluid membranes and predicting values for bending rigidity, compressibility, and even thermal expansion coefficient falling in the accepted range of values for bilayer membranes. But it still represents a generic membrane. Now, the authors have suggested a similar model for the archaeal bolalipids, which have chemically different lipids (the presence of cyclopentane rings for one), and there is no good justification for using the same pairwise interactions between their representative beads in the coarse-grained model. This does not necessarily diminish the worth of all the authors' analyses. What is at risk here is the confusion between "what we observe this model of bolalipid or mixed-membranes do" and "how real bolalipid-containing archaeal membranes behave at these mechanical and thermal conditions."*

As the reviewer correctly notes, Cooke and Deserno used a minimal model, devoid of chemical detail, to represent fluid lipid membranes composed of bilayer lipids. Indeed archaeal lipids are chemically different compared to non-archaeal lipids, but just like non-archaeal lipids, they can be very different from one another. Given the chemical diversity of bolalipids between each other, instead of representing their complexity in a complicated model with many experimentally unconstrained parameters, we here defined a minimal model for bolalipids. The power of this minimal model is to represent the key physical/geometrical characteristics of archaeal membranes, namely the fact that lipid heads on two sides of the membrane are often connected, that bolalipids can exhibit a conformational change, and that bolalipids mix with some percentage of bilayer molecules. We then ask a general question: how do these unique geometrical characteristics of archaeal membranes influence their mechanics and reshaping? The reviewer is however right in pointing out that a model, regardless of its level of details (atomistic, coarse-grained, minimal), is still a model.

Our approach of extending an established coarse-grained model for bilayer lipids to bolalipids is further supported by experimental observations, which report that archaeal bilayer lipids can form membranes of comparable bending rigidity to those of non-archaeal bilayer membranes [53]. Hence, different lipid linkages (archaeal vs. non-archaeal) give rise to fluid, deformable membranes of not too dissimilar rigidities, suggesting that both archaeal and non-archaeal bilayer lipids can be represented by a similar minimal coarse-grained model for the purpose of mesoscopic biophysical investigations. Since archaeal bolalipids have the same core chemical structure as two archaeal bilayer lipids joined by their tail ends, similarly we model a bolalipid by joining two bilayer lipids. Such an approach also efficiently enables us to compare bolalipid with bilayer membranes, and connect to the large body of knowledge on the physics of bilayer membranes.

To conclude, our coarse-grained model is indeed intended to capture the main physical properties of bolalipid membranes, and not their chemical diversity.

*R3.Q2: Another more specific, major issue has to do with using the Hamm-Kozlov model for fitting the power spectrum of thermal undulations. The  $1/q^2$  term can very well be attributed to membrane tension. While a barostat is indeed used, have the authors made absolutely sure that the deviation from  $1/q^4$  behaviour does not correspond to lateral tension?*

To the casual observer, any  $1/q^2$  trend might point at membrane tension. However, the precise functional form is relevant as it determines whether the  $1/q^2$  dominates the  $1/q^4$  trend for small or large values of the wave number  $q$  in the fitted power spectrum.

The first model (including lipid tilt) exhibits the functional form  $1/(kq^4) + 1/(kq^2)$ . In contrast, the second model (including membrane tension) exhibits the functional form  $1/(kq^4 + \Sigma q^2)$ . Importantly, the two models obey a different functional form. Here  $k$  and  $k_\theta$ , are the bending and tilt moduli, which are assumed positive, and  $\Sigma$  is the membrane tension, which can be either positive or negative. For the first model (with tilt), while for small  $q$  the amplitude is proportional to  $q^{-4}$ , for large  $q$  the amplitude is proportional to  $q^{-2}$ . In contrast, for the second model (with positive tension) while for small  $q$  the amplitude is proportional to  $q^{-2}$ , for large  $q$  the amplitude is proportional to  $q^{-4}$ . If membrane tension were to be negative in the second model, the slope would cross from negative infinity for small  $q$  to -4 for large  $q$ . The functional dependencies are summarized in Author response image 1A.

For rigid bolalipid membranes, it is clearly visible that the slope of the power spectrum plotted against the wave number  $q$  decreases with increasing  $q$  (Author response image 1B). While the slope initially assumes a value close to 4, it gradually approaches 2 for larger values of  $q$ . We conclude that only the model including lipid tilt can fit the power spectrum of membrane fluctuations appropriately (solid-dashed line), whereas the model with tension fails to fit the data (dashed line). We note that the combined model containing both lipid tilt and membrane tension does not give a better fit (dotted line).

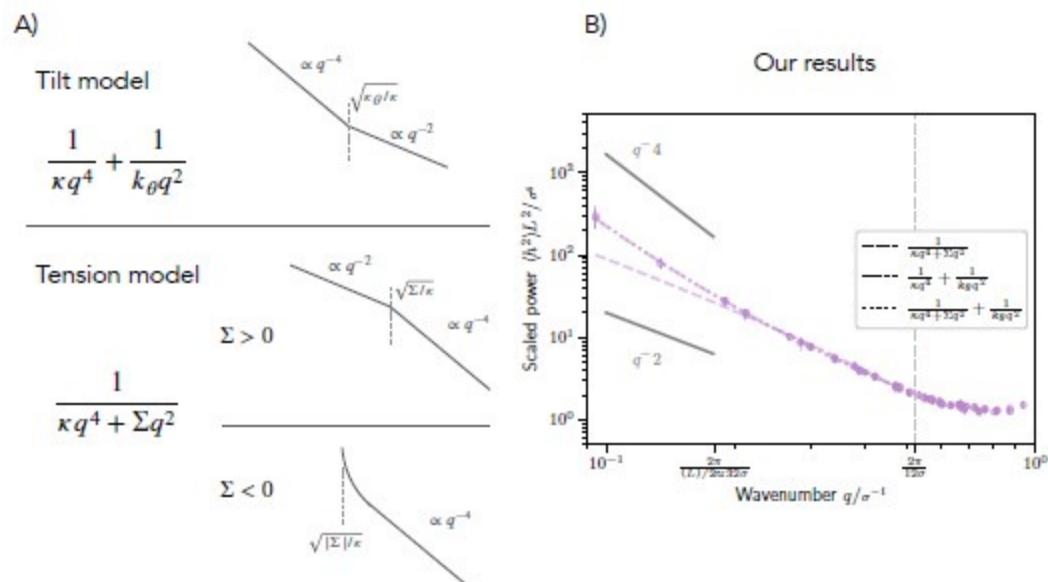
To demonstrate that the tension model cannot fit the data, we included the best fits for both models for rigid bolalipid membranes in the new SI section 16 (p. S22) and show that only the tilt model leads to acceptable fits. We also measured the projected membrane tension -

$\langle (P_x + P_y) L_z \rangle / 2$ , where  $P_x, P_y$  are respectively the pressure in  $x$  and  $y$  direction and  $L_z$  is the

dimension of the simulation box in  $z$  axis. We found the projected membrane tension to give a negligible value similarly to the one that we indirectly measured by fitting a combined model with both tension and tilt, further confirming our conjecture.

#### Author response image 1.

(A) Schematic showing the decay of the power spectrum as a function of the wave number  $q$  in the tilt model (top), in the tension model with positive membrane tension (middle), and in the tension model with negative membrane tension (bottom). (B) Fitted power spectrum as a function of  $q$  for rigid bolalipid membranes ( $k_{\text{bola}} = 5k_B T$ ). The fit shows that while the model with tension (dashed line) cannot fit the data, the model with tilt nicely fits the spectrum (solid-dashed line). The combined model including both tension and tilt does not fit the spectrum any better (dotted line).



R3.Q3: I got more worried when I noticed in the SI that the simulations had been done with combined "fix langevin" and "fix nph" LAMMPS commands. This combination does not result in a proper isothermal-isobaric ensemble. The importance of tilt terms for bolalipids is indeed very interesting, but I believe more care is needed to establish that.

In what follows, we show that there is no reason to worry. First of all we want to clarify that the physical setup we simulate is that of a membrane contained in a heat bath under negligible tension with correct diffusional dynamics. To achieve this physical setup, for which we use a Langevin thermostat combined with pressure control via an overdamped barostat, which we implement in LAMMPS by combining "fix langevin" and "fix nph".

In more detail: we simulated particles in an implicit solvent, for which we use a Langevin thermostat to get the right diffusional dynamics. To apply the theory of fitting fluctuation spectrums the simulation box length needs to be (near) constant. However, simulating membranes at a fixed box size results in an average non-zero membrane tension, making it hard to measure bending rigidity. The reason is that the effect of membrane tension is most influential on the largest wavelength modes, which are also most decisive when determining mechanical membrane properties like membrane rigidity. To minimize the effect of tension, we perform our simulation with an overdamped barostat ( $\tau_{\text{baro}} = 10 \tau_{\text{langevin}}$ ), which keeps the membrane near tensionless, as also done before [32]. In the revised manuscript, we have clarified the statement on the physical ensemble used (p.S2):

"For simulating flat membrane patches of bolalipids, we combined the previously used Langevin thermostat with relaxation time of  $1\tau$  with a Nosé-Hoover barostat with relaxation time of  $10\tau$ . In LAMMPS this amounts to combining the commands 'fix langevin' with 'fix nph'. We configured the barostat to set lateral pressure  $P_{xy}$  to zero by re-scaling the simulation box in the x-y plane. We compare this setup to a fixed box length setup, and an NPT ensemble setup, in SI section 17."

To connect our results with statistical mechanics ensemble theory we tested alternative setups. Similar setups, including the formal isothermal-isobaric ensemble, where  $N, P, T$  are kept constant using Nose-Hoover style equations for thermostating and barostating with modern corrections [34], which the reviewer refers to, result in very similar fluctuation spectrums. Consequently, our measurements of bending and tilt modulus hold true

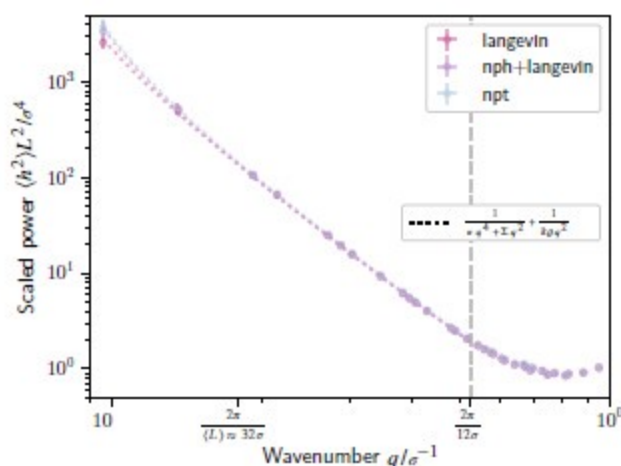
regardless of the integration scheme. However, such a setup does not correctly capture implicit solvent and diffusional dynamics.

In even more detail: we tested our setup (implemented via "fix langevin"+"fix nph") versus a isothermal-isobaric ensemble (implemented via "fix npt"). We measured volume mean and standard deviation, and found them matching for a reference LJ gas.

To be completely sure, and to please the reviewer, we have performed additional verifications in the new SI section 17, which we summarize in the following. We simulated three representative membranes with different integration schemes: "fix npt", "fix langevin"+"fix nph", and "fix langevin" (Langevin dynamics with projected area fixed at the average value obtained from a "langevin+nph"). We checked that the "fix nph" barostat is merely equilibrating the membrane to a tensionless configuration, after which the projected membrane area ( $A_p = L_x L_y$ ) is practically constant. Consequently, the different schemes resulted in minor changes in the longest wavelength modes that we tracked down to small changes in the negligible tension. The resulting measurements of bending modulus change by less than 10%, and our main text conclusions do not change. Author response image 2 compares the fluctuation spectrums for the different integration schemes.

### Author response image 2.

Height fluctuation spectrum, for a bilayer membrane at  $T_{\text{eff}} = 1.1$ , simulated with Langevin dynamics (pink, 'langevin'), our setup (purple, 'nph+langevin'), and under an isothermal-isobaric ensemble (blue, 'npt'); fits are shown as dotted lines.



*R3.Q4: This issue is reinforced when considering Figure 3B. These results suggest that increasing the fraction of regular lipids increases the tilt modulus, with the maximum value achieved for a normal Cooke-Deserno bilayer void of bolalipids. But this is contradictory. For these bilayers, we don't need the tilt modulus in the first place.*

We understand the concern why this might be counter-intuitive, and we thank the reviewer for pointing it out. We first want to stress that the tilt modulus can also be measured for bilayer membranes even if it is not needed to fit the fluctuation spectrum. If we measure the tilt modulus for a bilayer membrane, we obtain a value similar to the previously measured one [36]. Importantly, here we also report measurements for the tilt modulus for bolalipid membranes.

To understand the seemingly contradictory behaviour of the tilt modulus, it is insightful to rewrite the expression for the fluctuation spectrum as done in Eq. (1):

$$\begin{aligned}\langle |h_n|^2 \rangle &= \frac{k_B T}{L^2} \left( \frac{1}{\kappa q^4} + \frac{1}{\kappa_\theta q^2} \right) \\ &= \frac{k_B T}{L^2} \frac{1}{\kappa q^4} (1 + q^2 l_\theta^2),\end{aligned}$$

where  $l_\theta = \sqrt{\kappa/\kappa_\theta}$  is a characteristic length scale related to tilt, which we call the tilt

persistence length. From the last equation it is easy to see that the tilt modulus  $\kappa_\theta$  becomes relevant for the fluctuation spectrum if the tilt persistence length  $l_\theta$  is not negligible. In other words, this means that we have to consider the tilt modulus  $\kappa_\theta$  as relevant, if it is sufficiently small compared to the bending rigidity  $\kappa$ .

However, this is not only counter-intuitive, but also difficult to communicate graphically. Per the excellent reviewer's suggestion, to make the interpretation more accessible, we converted in the main text and its figures the tilt modulus to the more directly interpretable tilt persistence length  $l_\theta$ , as this is small when tilt is irrelevant (for bilayer lipids and flexible bolalipids) and large otherwise (for rigid bolalipids). This includes changes to the main text on p.6 and p.8, and to the insets in Figs. 2C and 3B. We note that for completeness we also report the tilt modulus  $\kappa_\theta$  in the SI.

*R3.Q5: Also, from the SI, I gathered that the authors have neglected the longest wavelength mode because it is not equilibrated. If this is indeed the case, it is a dangerous thing to do, because with a small membrane patch, this mode can very well change the general trend of the power spectrum. As a lot of other analyses in the manuscript rely on these measurements, I believe more elaboration is in order.*

We thank the reviewer for the careful examination of our supplementary material. For each fluctuation spectrum measurement, we ran multiple replicas. We observed that the largest wavelength modes were not fully equilibrated. In the simulations the first mode of the fluctuation spectrum is probed at different amplitudes and phases. We thus expected the potential systematic error would show up clearly when comparing spectrums of the different replicas. As we saw no correlation in these systematic offsets between replicas, we concluded that the simulations are sufficiently equilibrated and we could safely exclude the first mode of the fluctuation spectrum from our analysis.

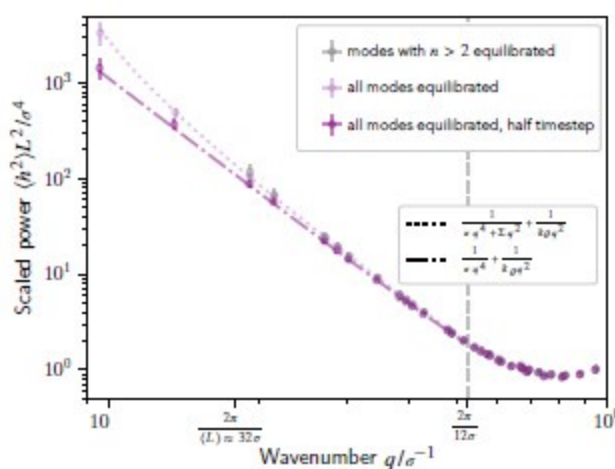
To show without doubt that this procedure does not randomly bias our results, we also ran simulations for three representative membranes until all modes were equilibrated. On the modes previously equilibrated, the resulting spectrums agree with our previous shorter simulations. On the largest wavelength modes that were previously not fully equilibrated, we noticed a small deviation from theory, specifically for flexible membranes (small bending modulus). These small deviations can be explained by including a negligible negative tension. Importantly, however, the resulting bending modulus  $\sigma$  stays nearly the same. We note that the small negative tension disappears when we halve the timestep (see Author response image 3). This verification is shown in SI section 17.

*R3.Q6: The authors have found that "there is a strong dependency of the bending rigidity on the membrane mean curvature of stiffer bolalipids." The effect is negative, with the membrane becoming less stiff at higher mean curvatures. Why is that? I would assume that with more flexible bolalipids, the possibility of reorganization into U-shaped chains should affect the bending rigidity more (as Figure 2E suggests). While for a stiff bolalipid, not much would change if you increase the mean curvature. This should be either a tilt effect, or have to do with asymmetry between the leaflets. But on the other hand, the tilt*

modulus is shown to decrease with increasing bolalipid rigidity. The authors get back to this issue only on page 10, when they consider U-shaped lipids in the inner and outer leaflets and write, "this suggested that an additional membrane-curving mechanism must be involved." But then again, in the Discussion, the authors write, "It is striking that membranes made from stiffer bolalipids showed a curvature-dependent bending modulus, which is a clear signature that bolalipid membranes exhibit plastic behaviour during membrane reshaping," adding to the confusion.

### Author response image 3.

Height fluctuation spectrum, for a bilayer membrane at  $T_{\text{eff}} = 1.1$ , as simulated in the main text (grey, for  $60 \rightarrow 10^3 \tau$ ), for longer duration ( $1_{.44} \rightarrow 10^6 \tau$ ) (pink), and with the longer duration and halved timestep  $= 0.005 \tau$  (purple); fits are shown as dotted lines (tension and tilt) or dash-dot lines (tilt only).



We thank the reviewer for asking this important question. Membrane bending rigidity in bolalipid membranes decreases dramatically once a small fraction of U-shapes is allowed to form, but then plateaus once this U-shape fraction reaches 20%. In a curved bolalipid membrane, U-shapes must accumulate in the outer leaflet to accommodate for area difference. Together, the bending rigidity non-linear dependence on U-shape fraction, and the promotion of U-shapes by curvature, explain why in a membrane made of moderately stiff bolalipids ( $k_{\text{bola}} = 1k_{\text{B}}T$ ), which contain very few U-shapes in the flat state, the bending rigidity of the membrane decreases as curvature increases. While in a membrane made of flexible bolalipid molecules ( $k_{\text{bola}} = 0$ ), where many U-shapes are present in the flat membrane, the bending rigidity does not change with curvature.

Bending rigidity  $\kappa$  in flat membranes composed of bolalipids decreases dramatically once a small fraction of U-shapes is allowed to form, but plateaus once more than 20% of U-shaped bolalipids are present. In details, our data shows that with an increasing bolalipid molecular rigidity  $k_{\text{bola}}$ , both the number of U-shaped bolalipids decreases (Fig. 2B) and the membrane rigidity  $\kappa$  increases (Fig. 2C). Thus, the correlation suggests that U-shaped bolalipids soften the membrane, in a non-linear way where most of the change in membrane bending rigidity happens for U-shaped bolalipid fraction  $< 20\%$  (Figure S11).

Separately, membrane curvature affects the area difference between curved membrane leaflets and thus drives U-shape accumulation. To be specific, a cylindrical membrane with area  $A$ , mean curvature  $H$  and thickness  $h$  has the outer leaflet with area  $A(1 + Hh)$  and the inner leaflet with smaller area  $A(1 - Hh)$ . This can be large, in our simulations up to an area change of  $Hh = 25\%$ . For pure bolalipid membranes, straight bolalipids occupy the same space

in each leaflet. Area difference can then be achieved only by having a different amount of U-shaped bolalipids in each leaflet, which can result in a different U-shape fraction between leaflets and thus ‘asymmetry between leaflets’. Figure S10 confirms U-shape head fraction asymmetry that increases with curvature, for both flexible ( $k_{\text{bola}} = 0$ ) and moderately stiff bolalipids ( $k_{\text{bola}} = 1k_{\text{B}}T$ ).

Together, these two effects result in membrane softening under curvature for the moderately stiff bolalipids, but constant rigidity for flexible bolalipids (Fig. 2F). In details: for membranes composed of moderately stiff bolalipid molecules ( $k_{\text{bola}} = 1k_{\text{B}}T$ ), the U-shape bolalipid head fraction only increases in the outer leaflet, going from 10 to 20% (Figure S10). This is in the high sensitivity region where the bending rigidity is expected to change the most (Figure S11). We hypothesize that the molecular rigidity of a U-shaped bolalipid creates compression on the outer leaflet that stabilizes the membrane curvature and thus causes membrane softening. We suspect that for membranes composed of rigid bolalipids ( $k > 1k_{\text{B}}T$ ), the effect is likely not present due to the absence of U-shape formation even under strong bending.

By contrast, for membranes composed of flexible bolalipids ( $k = 0$ ), the U-shaped bolalipid head fraction changes relatively little from its value for flat membranes (from 50% to respectively 60 and 40% for the outer and inner leaflet, Figure S10). This is in the region where the membrane bending rigidity is expected to respond weakly to U-shape fraction (Figure S11). Additionally, the change is symmetric, so presumably the outer leaflet becomes softer as the inner leaflet becomes stiffer, thus creating opposing effects and only weakly affecting the membrane bending rigidity as a whole. We note that the distinction between the U-shape head fraction that we plot (Figure S10) and U-shape fraction (Figure S11) matters little for this analysis.

We have added this deduction and its plots to SI section 8, and revised the corresponding statement in the main text accordingly (p.7).

“Changing membrane curvature alters the area differently in the two membrane leaflets. To adapt to the area difference, we thus expect the fraction of U-shaped bolalipids to change as the membrane curvature changes. Moreover, the results of Fig. 2B and Fig. 2C showed that the U-shaped bolalipid fraction and the membrane bending rigidity are correlated. As a result, we predict that the fraction of straight versus U-shaped bolalipids in a membrane will change in response to membrane bending, in a way that makes the bending rigidity of a bolalipid membrane curvature dependent.”

*R3.Q7: This issue is repeated when the authors study nanoparticle uptake. They write: “to reconcile these seemingly conflicting observations we reason that the bending rigidity, similar to Figure 2F, is not constant but softens upon increasing membrane curvature, due to dynamic change in the ratio between bolalipids in straight and U-shaped conformation. Hence, bolalipid membranes show stroking plastic behaviour as they soften during reshaping.” But the softening effect that they refer to, as shown in Figure 4B, occurs for very stiff bolalipids, for which not much switching to U-shaped conformation should occur.*

We thank the reviewer for locating a particularly dense sentence. We changed the text to explicitly refer to the range  $k \in [0, 2] k_{\text{B}}T$  for which there is significant change in U-shape fraction (p.8):

“To reconcile these seemingly conflicting observations we reason that the bending rigidity  $\kappa$ , similar to Fig. 2F, is not constant but softens in the range  $k \in [0, 2] k_{\text{B}}T$ , upon increasing membrane curvature. This is due to the dynamic change in the ratio between bolalipids in straight and U-shaped conformation.”

As for Fig. 4B, for  $k > 2k_{\text{B}}T$ , pores form thus explaining the plateau in adsorption energy.

*R3.Q8: Another major issue is with what the authors refer to as the “effective temperature”. While plotting phase diagrams for  $kT/\epsilon_p$  value is absolutely valid, I’m not a fan of calling this effective temperature. It is a dimensionless quantity that scales linearly with temperature, but is not a temperature. It is usually called a “reduced temperature”. Then the authors refer to their findings as studying the stability of archaeal membranes at high temperatures. I have to disagree because  $\epsilon_p$  is not the only potential parameter in the simulations (there are at least space exclusion and angle-bending stiffnesses) so one cannot identify changing  $\epsilon_p$  with changing the global simulation temperature. This only works when you have one potential parameter, like an LJ gas.*

We indeed thought about this before and found that it makes little difference in our set-up. To thoroughly show that the distinction matters very little, per reviewer’s question, we computed our phase diagrams by scaling temperature  $T$  explicitly (and not lipid tail interactions  $T_{\text{eff}} = k_B T / \epsilon_p$ ). We added these results to the SI section 14 and found no significant difference when comparing scaling tail interactions (Figure S15A) with scaling temperature explicitly (Figure S15B).

We also computed Fig. 2A-C for scaling interactions (Figure S17A) and scaling temperature explicitly (Figure S17B). We found a slightly increased U-shaped bolalipid fraction for low  $k$  when comparing scaling interactions (Figure S17A) with temperature scaling (Figure S17B). The reason is that the U-shaped fraction depends on temperature, as with higher temperature bolalipids can easier transition into the U-shape. Most importantly, however, we found no qualitative changes on the liquid region or the mechanical membrane properties when we compared the different scaling variants.

The reason why both scaling variants match so well can be understood easily. All pair potentials, including volume exclusion interactions between head beads and other membrane beads, were also scaled in the same manner as tail-to-tail interactions, as described in the SI. In contrast, the energy scales for maintaining the lipid bonds, the bilayer lipid angles and the bolalipid angles are relatively large compared to the energy scales involved in tail-to-tail interactions. This separation of energy scales guarantees that there will be little effect when increasing global temperature. Regarding nomenclature, we take the reviewer’s advice and have added ‘reduced temperature’ as an alias for  $T_{\text{eff}}$  in the main text.

In the revised version of the manuscript, we mention these observations in the SI section 14 and point towards these results in the main text (p.4):

“This interaction strength governs the membrane phase behaviour and can be interpreted as the effective temperature or reduced temperature  $T_{\text{eff}} = k_B T / \epsilon_p$ . As the distinction between scaling interactions ( $T_{\text{eff}}$ ) or temperature ( $T$ ) is not important for our analysis (see Supplemental Information (SI) section 14), for simplicity we refer to  $T_{\text{eff}}$  as temperature in the following.”

#### *Minor issues*

*R3.Q9: As the authors have noted, the fact that the membrane curvature can change the ratio of U-shaped to straight bolalipids would render the curvature elasticity non-linear (though the term “plastic” should not be used, as this is still structurally reversible when the stress is removed. Technically, it is hypoelastic behaviour, possibly with hysteresis.) With this in mind, when the authors use essentially linear elastic models for fluctuation analysis, they should make a comparison of maximum curvatures occurring in simulations with a range that causes significant changes in bolalipid conformational ratios.*

We thank the reviewer for their suggestion on calling the non-linear behaviour of the curvature elasticity hypoelastic. We have edited the main text accordingly (p.8):

“In an elastic material, the strain modulus holds constant and deformation is reversible. For bolalipid membranes at  $k = 1k_B T$ , however, the bending modulus decreases when deformation increases, rendering bolalipid membranes hypoelastic.”

Moreover, regarding the maximum curvatures occurring in the fluctuation simulations: We first note that the ensemble average of the mean curvature  $H$  from the fluctuation measurements is indicated as a vertical line in Fig. 2F. As the average value is nearly zero, the membrane can be considered as flat in good approximation. To investigate the question in more detail, we extended the SI with a careful analysis of the validity of the maximum membrane curvature and the validity of the Monge gauge approximation (SI section 15).

In short, we found that the involved membrane curvatures are small and therefore are unlikely to trigger any significant changes of the bending modulus. Moreover, since we are dealing with two bolalipid conformations, we also tested the homogeneity of the membrane. In our simulations of flat membrane patches we did not observe clustering or phase separation between the two bolalipid conformations beyond the  $[2,3]\sigma$  range. Furthermore, we get good agreement between our fluctuation measurement and the cylinder simulations in Fig. 2F. We now mention this verification in the revised version of the manuscript (p.8):

“Fortunately, this dependency on curvature does not invalidate our fluctuation results, where the curvature is small enough that its effect on the bending modulus is negligible (SI section 15).”

Last but not least, simulating bending/unbending cycles of an arc-shaped membrane (frozen endpoints) shows agreement with cylinder membrane simulations, and no hysteresis at the rates of deformation employed (cf. M. Amaral’s thesis [54], soon to be out of the embargo period).

*R3.Q10: The Introduction section of the manuscript is written with a biochemical approach, with very minor attention to the simulation works on this system. Some molecular dynamics works are only cited as existing previous work, without mentioning what has already been studied in archaeal membranes. While some information, like the binding of ESCRT proteins to archaeal membranes, though interesting, helps little to place the study within the discipline. The Introduction should be revised to show what has already been studied with simulations (as the authors mention in the Discussion) and how the presented research complements it.*

The present research for the first time covers archaeal membranes with a single coarse-grained model capable of assuming both bolalipid in-membrane conformations and sweeps through temperature, membrane composition, and molecular rigidity. The work shows the first curvature dependent bending modulus for pure bolalipid membranes. It also investigates systematically bending modulus and Gaussian modulus, and tests the model in an all-encompassing budding simulation that incorporates topology changes. Existing atomistic or coarse-grained MD simulations (MARTINI or similar force fields) are limited to small patches of membrane, with no study of large-scale deformations or topology changes; plus, they rely on force fields that were parametrized for bilayer membranes.

To give a comprehensive overview of the field, we revised the introduction section of the manuscript, in which we now discuss previous computational work investigating membrane diffusivity, U-shaped lipid fraction, and bending rigidity (p.3):

“By contrast, only a few studies have investigated bolalipid membranes applying computational or theoretical tools [24, 25]. Specifically, the pore closure time in bolalipid

membranes, and the role of cyclopentane rings for membrane properties has been investigated using all-atom simulations, showing decreased lateral mobility, reduced permeability to water, and increased lipid packing [26–28]. Moreover, using coarse-grained simulations, it was suggested that bolalipid membranes are thicker [29], exhibit a gel-to-liquid phase transition at higher temperature [30], and exhibit a reduced diffusivity [31]. However, little research has been devoted to investigating mechanics and reshaping of bolalipid membranes at the mesoscale despite the obvious importance of this question from evolutionary, biophysics, and biotechnological perspectives and although different membrane physics is expected to manifest.”

Following the reviewer’s advice and to keep the introduction concise and focused on bolalipid membranes, we have removed the paragraph on ESCRT-III proteins in the revised manuscript.

*R3.Q11: The authors have been a bit loose with using the term “stability”. I’d like to see the distinction in each case, as in “chemical/thermal/mechanical/conformational stability”.*

We have clarified when applicable the type of stability throughout the manuscript. In all other instances, if not clear from context, we mean simply that the membrane persists being a membrane. At our coarse-grained level, this means the membrane does not disassemble into a gas phase.

*R3.Q12: In the original Cooke-Deserno model, a so-called “poorman’s angle-bending term” is used, which is essentially a bond-stretching term between the first and third particle. However, I notice the authors using the full harmonic angle-bending potential. This should be mentioned.*

This is made clear in the SI (Eq. (S3)). Cooke and Deserno mention the harmonic angle potential as a valid alternative in their original publication. We now also added this detail to the main text (p.3):

“The angle formed by the chain of three beads is kept near 180° via an angular potential with strength  $k_0$ , instead of the approximation by a bond between end beads of the original model [32].”

*R3.Q13: The analysis of energy of U-shaped lipids with the linear model  $E = c_0 + c_1k$  is indeed very interesting. I am curious, can this also be corroborated with mean energy measurements? The minor issue is calling the source of the favorability of U-shaped lipids “entropic”, while clearly an energetic contribution is found. The two conformations, for example, might differ in the interactions with the neighbouring lipids.*

We were also curious and thank the reviewer for the suggestion of mean energy measurements. We concluded that there must be either an entropic contribution to the free energy or an intermolecular interaction energy favouring U-shaped bolalipids. We have now included these measurements in SI section 6 (p.S5):

“By splitting the average potential energy between an internal contribution (bonds, angles and pair interactions between particles in the same molecule) and an external contribution (pair interactions between a molecule and its neighbours), we determined the transition energy from straight to U-shaped bolalipids in detail. We found that this transition lowers the internal potential energy of the bolalipid while increasing its interaction energy. In total, we obtained an energy barrier for the transition of  $\Delta E_{s \rightarrow u} = 0.79 \pm 0.01 k_B T$ . Since the fit indicates, however, that the U-shaped bolalipid conformation is preferred over the straight

conformation, we conclude that there must be either an entropic contribution to the free energy or an intermolecular interaction energy favouring U-shaped bolalipids.”

We refer to these measurements in the main text (p.6 ):

“For the fit it appears that  $c_0 < 0$ , which implies that bolalipids in U-shape conformation are slightly favoured over straight bolalipids at  $k = 0$  (explored in SI section 6).”

*R3.Q14: The authors write in the Discussion, “In any case, our results indicate that membrane remodelling, such as membrane fission during membrane traffic, is much more difficult in bolalipid membranes [34].” Firstly, I’m not sure if studying the dependence of budding behaviour on adhesion energy with nanoparticles is enough to make claims about membrane fission. Secondly, why is the 2015 paper by Markus Deserno cited here?*

We thank the reviewer for giving us the opportunity to clarify. We make an energetic argument on membrane fission based on the observed difference in the ratio of  $|\bar{\kappa}|/\kappa$ .

Splitting a spherical membrane vesicle into two spherical vesicles (fission) increases the bending energy by  $8\pi\kappa$  and decreases the energy related to the Gaussian bending modulus by  $4\pi|\bar{\kappa}|$ . The second part of the argument is given for example in the review by Markus Deserno (p.23, right column), that’s why we cite the paper here. Together, this gives an energy barrier, required for membrane fission in the considered geometry of  $\Delta E_{\text{fission}} = 4\pi\kappa(2 - |\bar{\kappa}|/\kappa)$ . We

found that  $|\bar{\kappa}|/\kappa$  is around 0.5 for bolalipid membranes and around 1 for bilayer

membranes. Since  $\kappa$  was typically larger in bolalipid membranes we thus expect the energy barrier for fission  $\Delta E_{\text{fission}}$  to be larger for bolalipid membranes. We therefore predict that membrane remodelling, such as membrane fission during membrane trafficking, is harder in bolalipid membranes. We explain our reasoning in the discussion of the revised manuscript (p.13 ):

“Membrane remodelling, such as the fission of one spherical vesicle into two, increases the bending energy by  $8\pi\kappa$  but decreases the energy related to the Gaussian modulus by  $-4\pi|\bar{\kappa}|$

[39], giving rise to a fission energy barrier of  $\Delta E_{\text{fission}} = 4\pi\kappa(2 - |\bar{\kappa}|/\kappa)$ . Our results indicated

that while in bolalipid membranes  $\kappa$  is larger,  $|\bar{\kappa}|/\kappa$  is smaller compared to bilayer

membranes. Our results thus predict a larger energy barrier for membrane fission  $\Delta E_{\text{fission}}$  in bolalipid membranes compared to bilayer membranes.”

*R3.Q15: In the SI, where the measurement of the diffusion coefficient is discussed, the expression for  $D$  is missing the power 2 of displacement.*

We thank the reviewer for spotting this oversight. We corrected it in the revised version of the SI (p.S5 ).

*R3.Q16: Where cargo uptake is discussed, the term "adsorption energy" is used. I think the more appropriate term would be "adhesion energy".*

For the sake of simplicity, we changed the term to adhesion energy (caption of Fig. 4, and p.10). We do not have a strong opinion on this, but we believe that adsorption energy would be equally correct as we describe the adsorption of many lipid head beads to a nanoparticle.

*R3.Q17: Typos:*

*Page 1, paragraph 2: Adaption → Adaptation. Page 10, paragraph 1: Stroking → Striking.*

We thank the reviewer for spotting these typos which we have corrected in the revised version of the manuscript.

### **Recommendations for the authors**

#### **Reviewer #1 (Recommendations for the authors):**

*A few thoughts (likely out of the scope of this paper but possibly to consider upon revision):*

*R1.Q1: Do bolalipids always have the same headgroup? I don't recall reading this in the introduction/discussion. R1 and R2 are in Figure 1, but I don't know whether there are standard types. Could this be expanded upon? Is the model able to take these differences into account?*

We thank the reviewer for raising this important question. Similar to bacteria and eukaryotes, in archaea there is a huge variety in terms of the different head groups that lipids can contain and thus also lipid variety. Most archaeal lipids have head groups that contain either phosphate groups or sugar residues. Typically, archaeal bolalipids are asymmetric and contain a phosphatidyl and a sugar moiety at the two ends of the lipid molecule. Within the membrane the lipid is oriented such that the phosphatidyl moiety points towards the interior of the cell whereas the sugar moiety points towards the outside of the cell as it occupies more space [5].

In our computational model, however, we consider symmetric bolalipids for the sake of simplicity and to decouple the role of "connected geometry" from other effects. In principle, we could investigate the effect of lipid asymmetry by increasing the size of one of the lipid head beads. However, this investigation exceeds the scope of the present study and therefore requires future work.

In the revised version of the manuscript, we now clarify that bolalipids can have different headgroups (p.1 and the caption of Fig. 1):

"The hydrophilic heads can be composed of different functional groups with phosphatidyl and sugar being the most relevant moieties. For bolalipids the two head groups at either end of the molecule are typically distinct (Fig. 1A right) [5]."

"The hydrophilic head of a bolalipid can be composed of different functional groups represented by R1 and R2 (right)."

We also explicitly state that we neglect lipid head group asymmetry for the sake of simplicity (p.4):

“To decouple the effect of the connected geometry of the bolalipids from that of lipid asymmetry, we assume both head beads of a bolalipid to share the same properties.”

*R1.Q2: Is it possible to compare the mesoscale models to either Coarse-grained or even all-atom lipid models? Have simulations previously been performed for bolalipids at those levels of description?*

A few studies have investigated bolalipids membranes in simulations previously. These studies either used all-atom or coarse-grained simulations. However, none of these studies investigated how bolalipids respond to membrane deformations. Therefore, it is currently not possible to directly compare our results to studies in the literature. However, to recapitulate our predictions experimentally is certainly something that could and should be done in the future. As a reply to this reviewer and reviewer 3, we discuss the current state of modelling bolalipid membranes in simulations in the revised version of the manuscript (p.3):

“By contrast, only a few studies have investigated bolalipid membranes applying computational or theoretical tools [24, 25]. Specifically, the pore closure time in bolalipid membranes, and the role of cyclopentane rings for membrane properties has been investigated using all-atom simulations, showing decreased lateral mobility, reduced permeability to water, and increased lipid packing [26–28]. Moreover, using coarse-grained simulations, it was suggested that bolalipid membranes are thicker [29], exhibit a gel-to-liquid phase transition at higher temperature [30], and exhibit a reduced diffusivity [31]. However, little research has been devoted to investigating mechanics and reshaping of bolalipid membranes at the mesoscale despite the obvious importance of this question from evolutionary, biophysics, and biotechnological perspectives and although different membrane physics is expected to manifest.”

We want to mention, however, that we do compare membrane diffusivity, U-shaped lipid fraction, and bending rigidity to the behaviour and values that have been previously measured in simulations in the discussion section. In general, we find good agreement between our results and previously reported behaviour/values (p.13):

“While flexible bolalipid membranes are liquid under the same conditions as bilayer membranes, we found that stiff bolalipids form membranes that operate in the liquid regime at higher temperatures. These results agree well with previous molecular dynamics simulations that suggested that bolalipid membranes are more ordered and have a reduced diffusivity compared to bilayer membranes [24, 29]. In our simulations, this is due to the fact that completely flexible bolalipids molecules adopt both straight (transmembrane) as well as the U-shaped (loop) conformation with approximately the same frequency. In contrast, stiff bolalipids typically only take on the straight conformation when assembled in a membrane. These results agree with the previous coarse-grained molecular dynamics simulations using the MARTINI force field which showed that the ratio of straight to U-shaped bolalipids increased upon stiffening the linker between the lipid tails [29].

[...]

When we determined the bending rigidity of bolalipid membranes by measuring their response to thermal fluctuations, we found that membranes made from flexible bolalipids are only slightly more rigid than bilayer membranes. This result is consistent with previous atomistic simulations, which showed that the membrane rigidity was similar for membranes composed of bilayer lipids and flexible synthetic bolalipids [45].”

*R1.Q3: How would membrane proteins alter the behaviour of bolalipids? Either those integral to the membrane or those binding peripherally?*

The reviewer asks an important question. However, the question is difficult to answer due to its scope and the gaps in the current literature. Important examples of integral or peripheral membrane proteins that alter the behaviour of bolalipids and archaeal bolalipid membranes are involved in cell homeostasis, cell division, membrane trafficking, and lipid synthesis.

The cells of many archaeal species are enclosed in a paracrystalline protein layer called the Slayer, which is attached to the lipid membrane [4, 55]. The main function of the S-layer is to keep the cell's shape and to protect it against osmotic stress. Due to the embedding of the S-layer in the membrane at specific locations, it is to be expected that the membrane properties are influenced by the S-layer. Furthermore, archaea execute cell division by locally reshaping the membrane using FtsZ and ESCRT-III proteins [56]. While Asgard archaeal genomes encode proteins with homology to those regulating aspects of eukaryotic membrane remodelling and trafficking [57], they have yet to be observed undergoing a process like endocytosis [58]. In addition, it has been speculated that the proteins that drive the synthesis of two diether lipids into a tetraether lipid are either membrane associated or integral membrane proteins [59].

However, to the best of our knowledge it is not known how membrane proteins specifically alter the behaviour of bolalipids. Future work will need to be executed to answer this question. Following the advice of reviewer 3 and to keep the introduction concise and focused on bolalipid membranes, we do not mention these observations in the revised manuscript.

*R1.Q4: Is there a mechanism in cells to convert or switch bolalipids from a straight to a u-shaped description? Does this happen spontaneously or are there enzymes responsible for this?*

We thank the reviewer for bringing up this important point. Despite the relevance of the question, little is currently known about the mechanism that make bolalipids transition between a straight and a U-shaped configuration mainly because there is to date no established experimental method.

Besides our own results, most of what we know comes from coarse-grained molecular dynamics simulations, which showed that bolalipids can spontaneously transition between the straight and U-shaped configuration [29]. In addition, by using comparative genomic analysis, it has been predicted that many archaeal species contain flippases, i.e., membrane proteins that are able, upon the consumption of energy, to transfer (flipflop) bilayer lipids between the two membrane leaflets [43]. Moreover, it has been shown that *Halobacterium salinarum* (an archaeon with a bilayer lipid membrane) [44] contains scramblases, which are membrane proteins that passively transfer bilayer lipids from one membrane leaflet to the other. It is therefore tempting to speculate that similar proteins might exist for bolalipids which could facilitate the straight to U-shaped transition.

In addition, it has been reported that vesicles composed of bolalipid membranes can undergo fusion with enveloped influenza viruses [17]. In this context, it has been suggested that the influenza fusion protein hemagglutinin may locally induce U-shaped bolalipids to facilitate membrane fusion. However, all these hints are by far no proof of a mechanism that can drive the straight to U-shaped bolalipid transition, and further work needs to be done to investigate this question in detail.

In the revised version of the manuscript, we now discuss what is known about potential mechanisms to facilitate the straight to U-shaped transition in the discussion section (p.13):

“While previous coarse-grained simulations predicted that bolalipids spontaneously transition between the straight and U-shaped conformations [29], how this happens in

archaeal membranes and whether membrane proteins are involved in this conformational transition needs to be clarified in the future. Experimental studies suggest that archaeal membranes contain flippases and scramblases for the transitioning of bilayer lipids between membrane leaflets [43, 44], raising the possibility that similar proteins could also facilitate conformational transitions in bolalipids. In addition, it has been suggested that the viral fusion protein hemagglutinin could cause a transition from straight to U-shaped bolalipid conformation during the fusion of bolalipid vesicles with influenza viruses [17]. However, future investigation is required.”

*R1.Q5: Ideally, coordinates and any parameter files required to run the molecular simulations should be included for reproducibility.*

We absolutely share the reviewer’s concern with reproducibility and as such have included in the original submission as part of our data availability section a link to a code repository (available at: <https://doi.org/10.5281/zenodo.13934991> [51]) that allows initializing and simulating flat membrane patches, with user control of the parameters explored in this paper ( $\omega, T_{\text{eff}}, k_{\text{bola}}, f^{\text{bi}}$ ).

**Reviewer #2 (Recommendations for the authors):**

*This is a great paper and I congratulate the authors for writing such a fine piece of scholarship. The only nitty-gritty feedback that I have is summarized in the following three points:*

*R2.Q1: In the introduction the authors talk about archaea adapting their membrane to retain membrane fluidity. However, homeoviscous adaptation is also fundamental in bacteria and eukaryotes.*

The reviewer is correct, like archaea the membranes of bacteria and eukaryotes must balance between flexibility and stability. Moreover, the cell membranes in all 3 domains of life need to maintain membrane fluidity and provide mobility to the embedded lipids and membrane proteins (homeoviscous adaptation). The general idea is that these organisms change the ratio of different lipids to change membrane properties and thereby optimally adapt to their environments [10]. Importantly, however, there are differences of how homeoviscous adaptation is maintained across the different domains of life. As a reply to this reviewer and reviewer 3, we now discuss the underlying mechanisms in the revised parts of the introduction (p.1):

“Like for bacteria and eukaryotes, archaea must keep their lipid membranes in a fluid state (homeoviscous adaptation). This is important even under extreme environmental conditions, such as hot and cold temperatures, or high and low pH values [7]. Because of this, many archaea adapt to changes in their environment by tuning the lipid composition of their membranes: altering the ratio between bola- and bilayer lipids in their membranes [8, 9] and/or by changing the number of cyclopentane rings in their lipid tails, which are believed to make lipid molecules more rigid [5]. For example, *Thermococcus kodakarensis* increases its tetraether bolalipid ratio from around 50% to over 80% when the temperature of the environment increases from 60 to 85 C [10]. Along the same lines, the cell membrane of *Sulfolobus acidocaldarius*, can contain over 90 % of bolalipids with up to 8 cyclopentane rings at 70 C and pH 2.5 [5, 11]. It is worth mentioning that in exceptional cases bacteria also synthesise bolalipids in response to high temperatures [12], highlighting that the study of bolalipid membranes is relevant not only for archaeal biology but also from a general membrane biophysics perspective.”

*R2.Q2: Uncertainties in Gaussian rigidity modulus estimates are not properly reported.*

The large uncertainties in the Gaussian rigidity modulus  $\bar{\kappa}$  were due to the fact how they were calculated. In short,  $\bar{\kappa}$  is determined in cap folding simulations [41] (SI section 9), by using the measured values of the dimensionless parameter  $\xi$ , related to the folding probability, the bending modulus  $\kappa$ , the membrane line tension  $\sigma$ , and the cap radius  $R$ . In our case, the main source of uncertainty for determining  $\bar{\kappa}$  comes from the uncertainty in the measurement of the bending rigidity  $\kappa$ . To obtain  $\kappa$ , previously, we fitted fluctuation spectra for different seeds and only then averaged the obtained values. In the revised version of the manuscript, we now first pool the fluctuation spectra of the different simulation seeds before we fit all spectra at the same time. This new approach results in smaller uncertainties for the bending rigidity  $\kappa$  and also the Gaussian rigidity modulus  $\bar{\kappa}$ .

As a consistency check, in addition to the simulations that we previously performed at  $T_{\text{eff}} = 1.3$ , we have repeated the cap folding and line tension simulations at  $T_{\text{eff}} = 1.2$ , resulting in similar values for  $\bar{\kappa}$ . In the revised version of the manuscript, we report the newly calculated values and uncertainties for  $\bar{\kappa}$  at  $T_{\text{eff}} = 1.2$  in the main text (p.8):

“At  $T_{\text{eff}} = 1.2$ , we obtained  $\bar{\kappa} = 4.30 \pm 0.22 \text{ kBT}$  and thus a ratio of  $\bar{\kappa}/\kappa = 0.89 \pm 0.04$  for bilayer membranes, similar to what has been reported previously [41]. For flexible bolalipid membranes, we got a slightly smaller value for  $\bar{\kappa} = 5.04 \pm 0.37 \text{ kBT}$ . Due to the larger bending modulus, however, flexible bolalipid membranes show a significantly smaller ratio  $\bar{\kappa}/\kappa = 0.64 \pm 0.04$  ( $k = 0$ ). At larger temperature ( $T_{\text{eff}} = 1.3$ ), the ratio can be even smaller  $\bar{\kappa}/\kappa = 0.45 \pm 0.07$  (see SI section 9).”

In addition, we report the values at  $T_{\text{eff}} = 1.3$  and  $T_{\text{eff}} = 1.2$  in the SI (p.S15, Tabl. S4):

We have also adapted the discussion of the Gaussian bending modulus accordingly (p.13):

“Another marked difference between bilayer and flexible bolalipid membranes is the ratio of the Gaussian rigidity to the bending modulus. Instead of being around 1 as for bilayer membranes [41], it is around 1/2 and therefore only half of that of bilayer lipids.”

**Reviewer #3 (Recommendations for the authors):**

*While I think the bulk of the work presented is useful, some of the issues that I raised in my review are indeed major. Without properly addressing them, it is hard to accept the conclusions of the manuscript. I hope the authors can address them by revising their analysis.*

We thank the reviewer for their constructive feedback, which helped us to improve the manuscript. We have addressed all points raised by the reviewer in our detailed point-by-point response to the reviewer (see above). We hope the reviewer will now find it easier to accept our conclusions.

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